

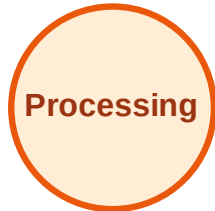
BFS Traversal

Step 1: Start at A

Legend



Complete



Processing



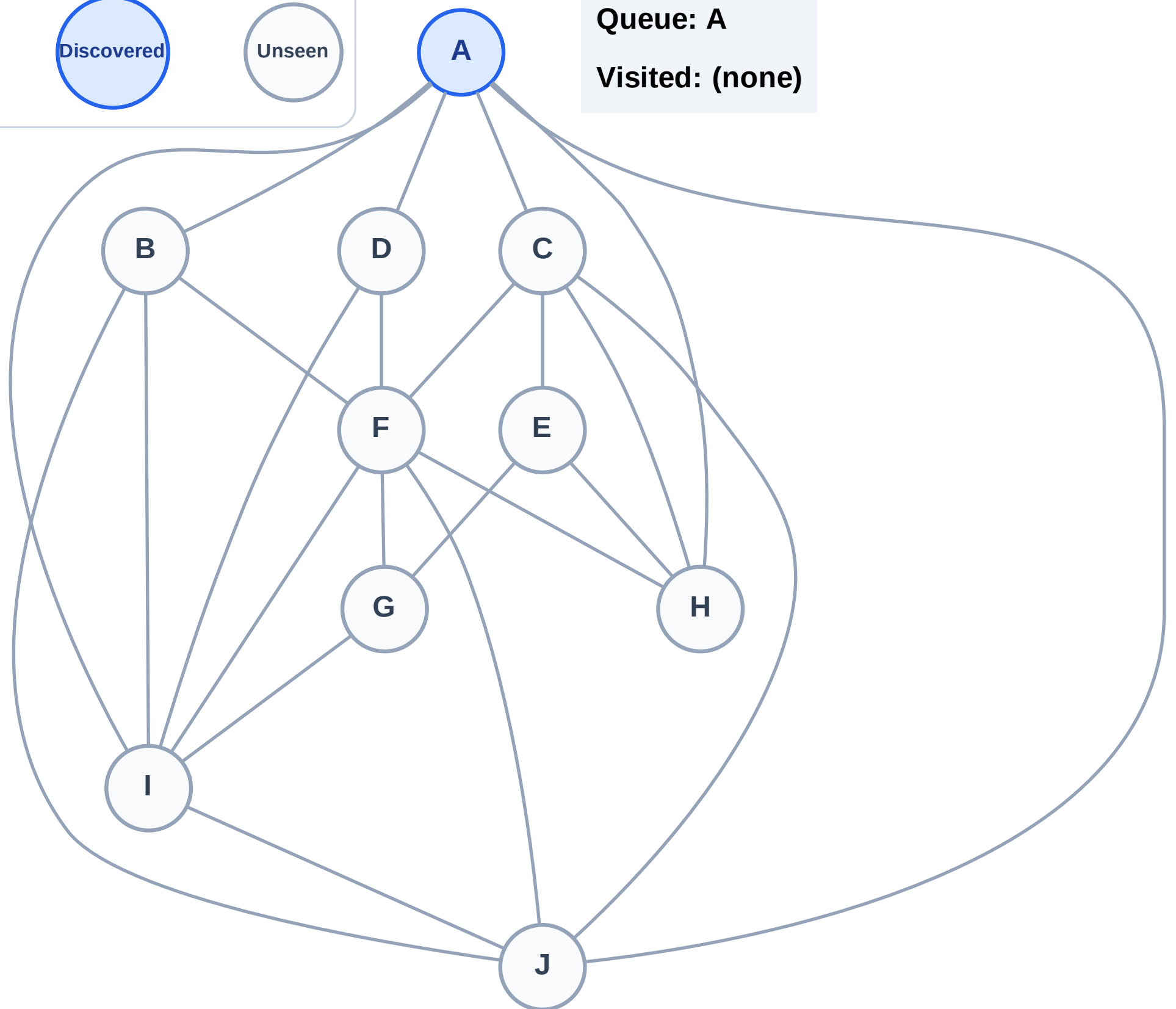
Discovered



Unseen

Queue: A

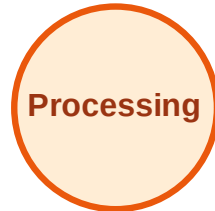
Visited: (none)



BFS Traversal

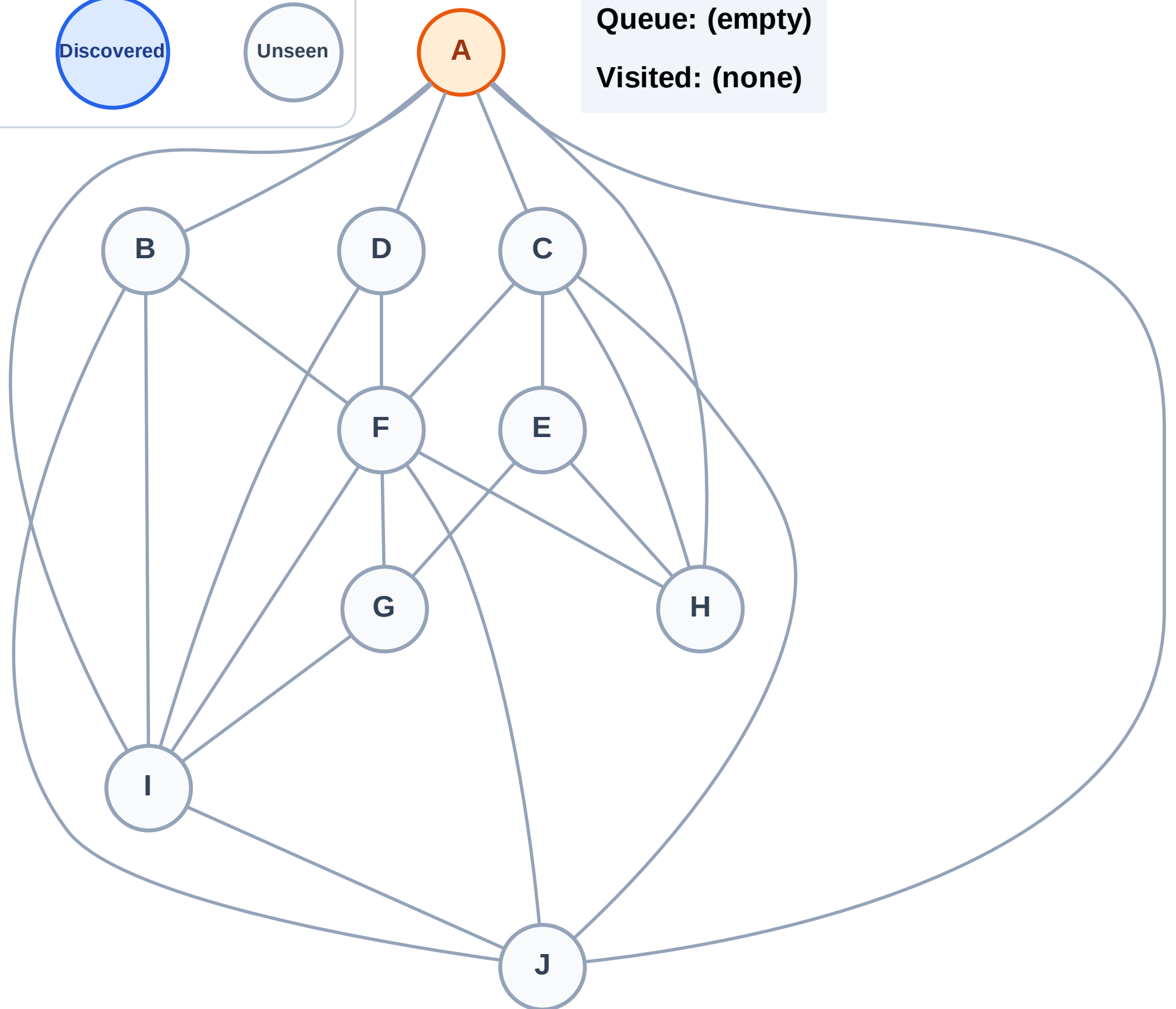
Step 2: Process node A

Legend



Queue: (empty)

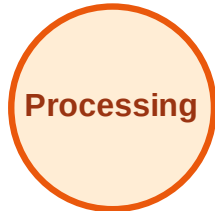
Visited: (none)



BFS Traversal

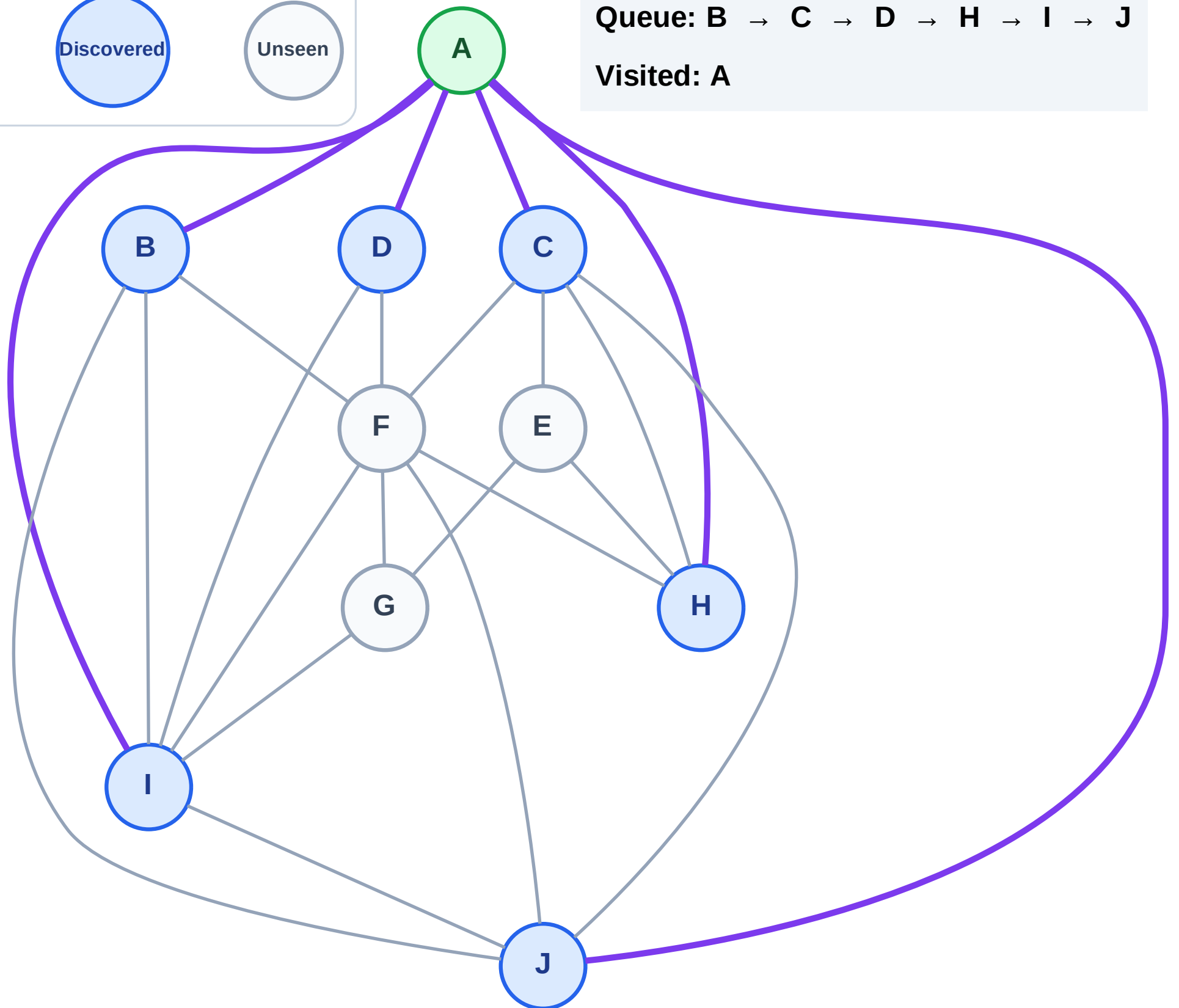
Step 3: Discover B, C, D, H, I, J from A

Legend



Queue: B → C → D → H → I → J

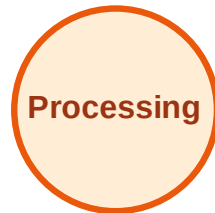
Visited: A



BFS Traversal

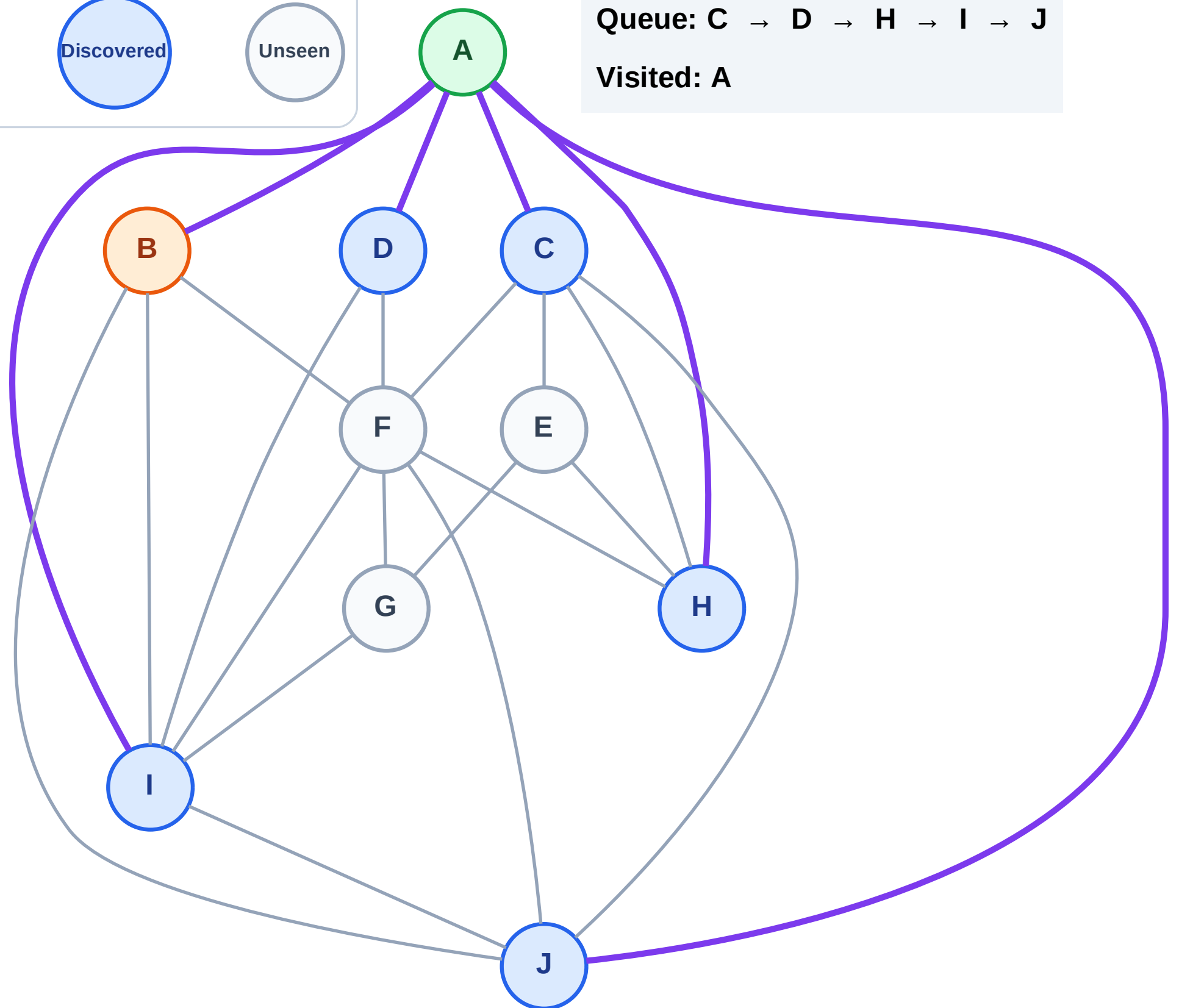
Step 4: Process node B

Legend



Queue: C → D → H → I → J

Visited: A



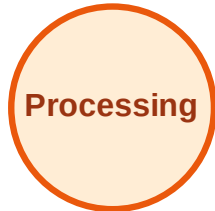
BFS Traversal

Step 5: Discover F from B

Legend



Complete



Processing



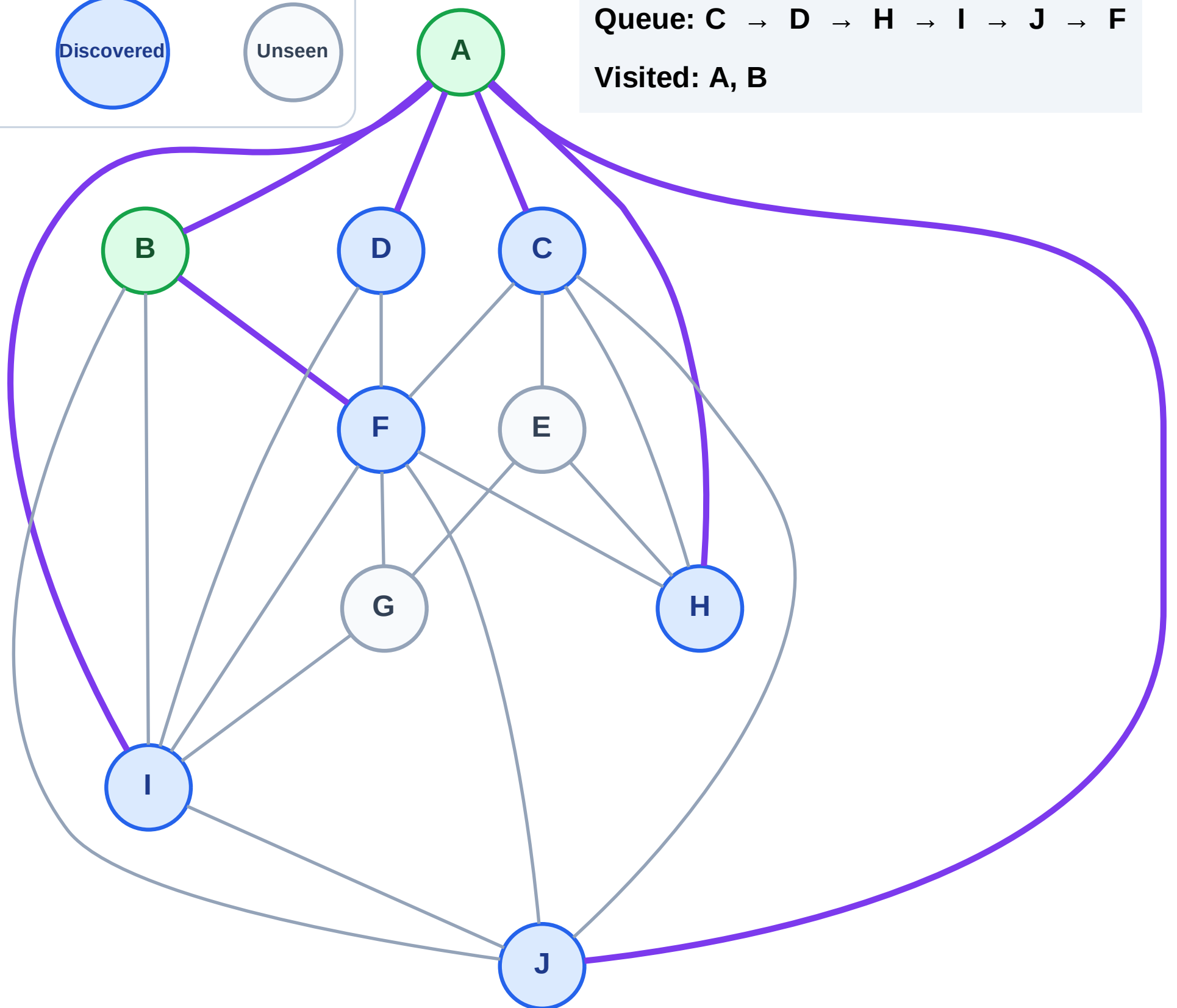
Discovered



Unseen

Queue: C → D → H → I → J → F

Visited: A, B



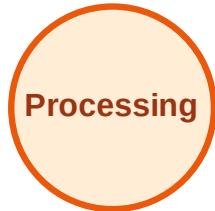
BFS Traversal

Step 6: Process node C

Legend



Complete



Processing



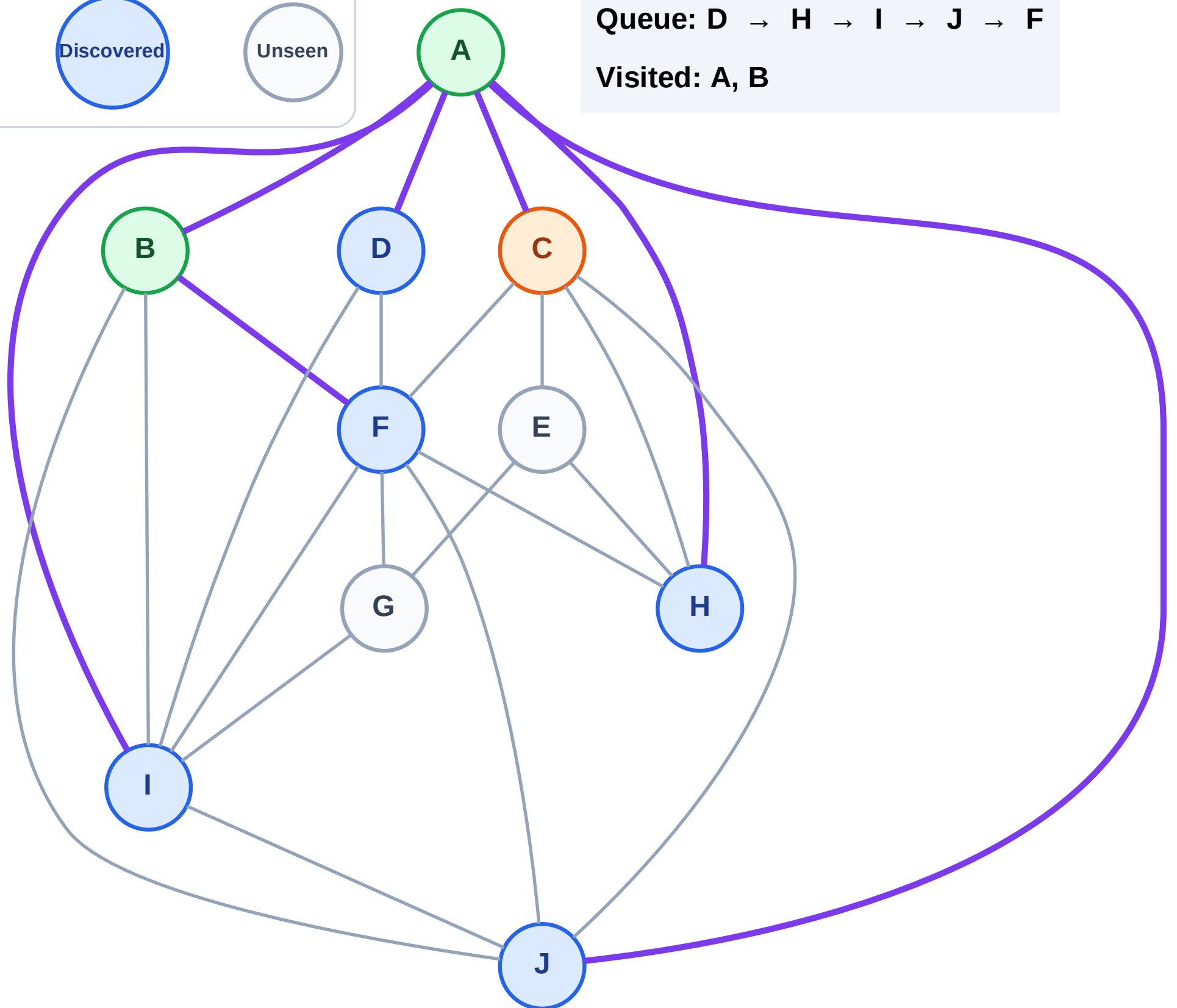
Discovered



Unseen

Queue: D → H → I → J → F

Visited: A, B



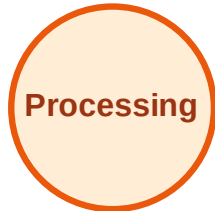
BFS Traversal

Step 7: Discover E from C

Legend



Complete



Processing



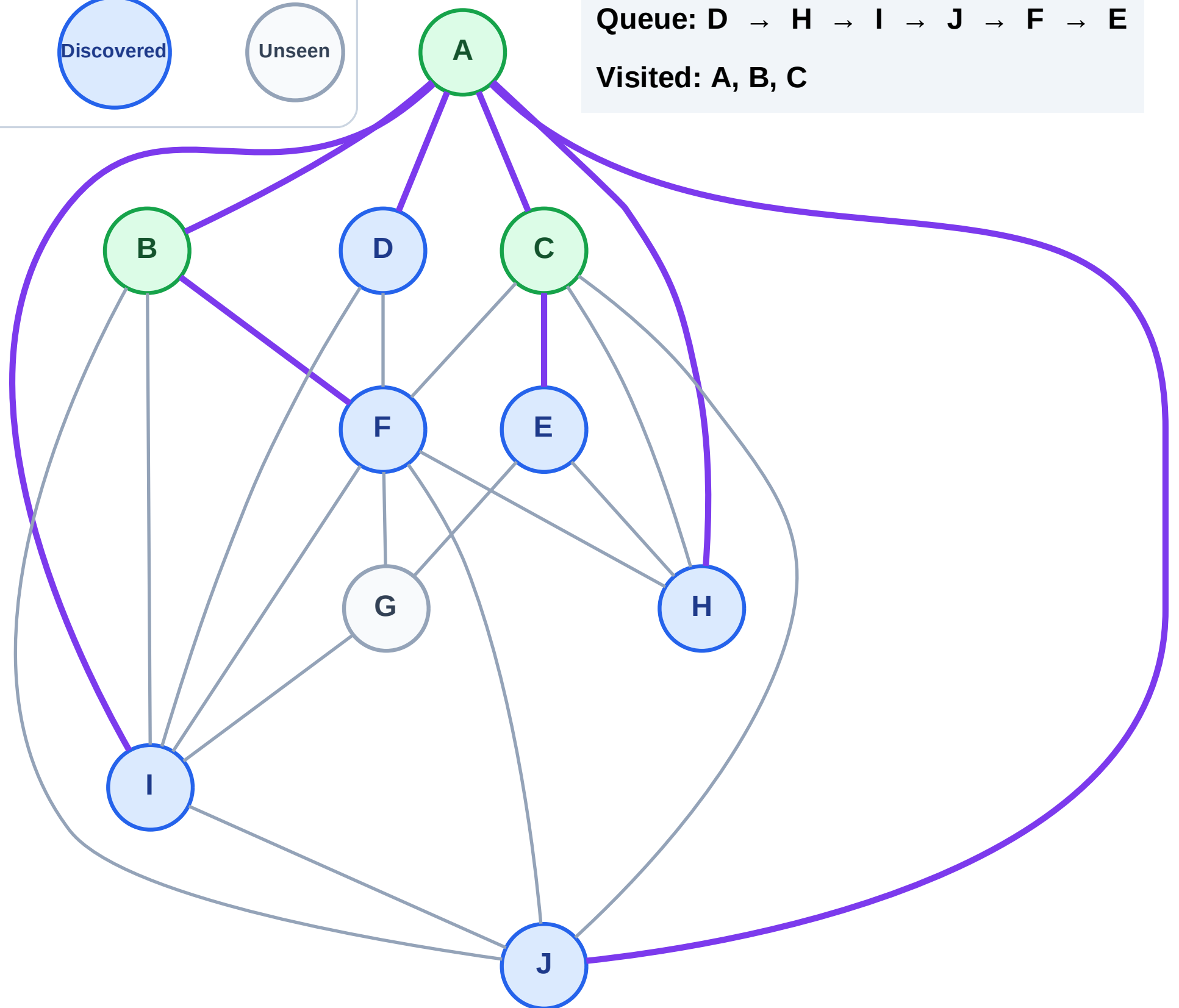
Discovered



Unseen

Queue: D → H → I → J → F → E

Visited: A, B, C



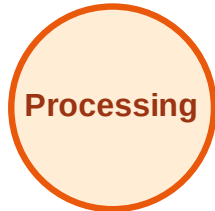
BFS Traversal

Step 8: Process node D

Legend



Complete



Processing



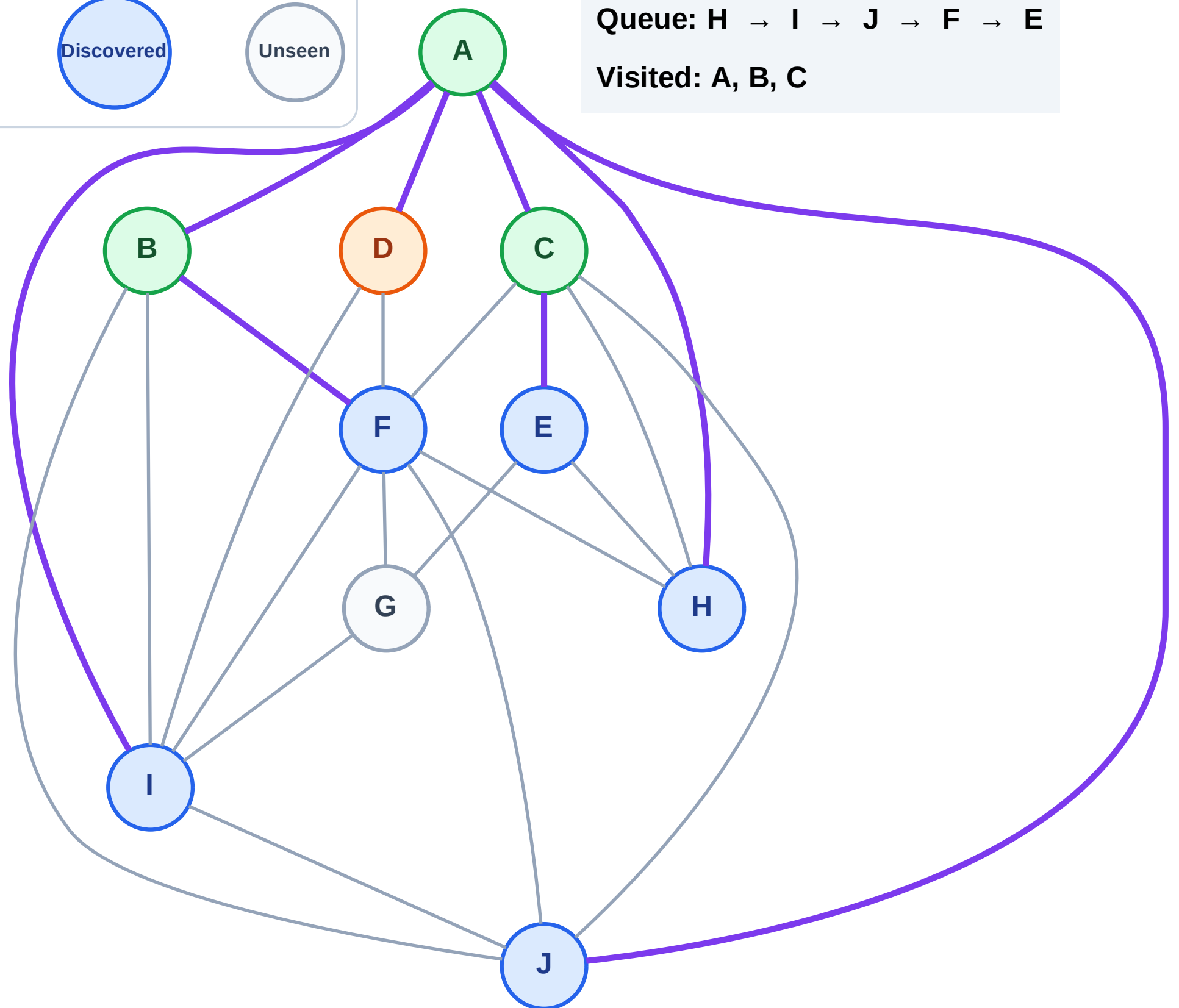
Discovered



Unseen

Queue: H → I → J → F → E

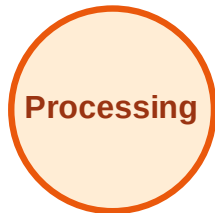
Visited: A, B, C



BFS Traversal

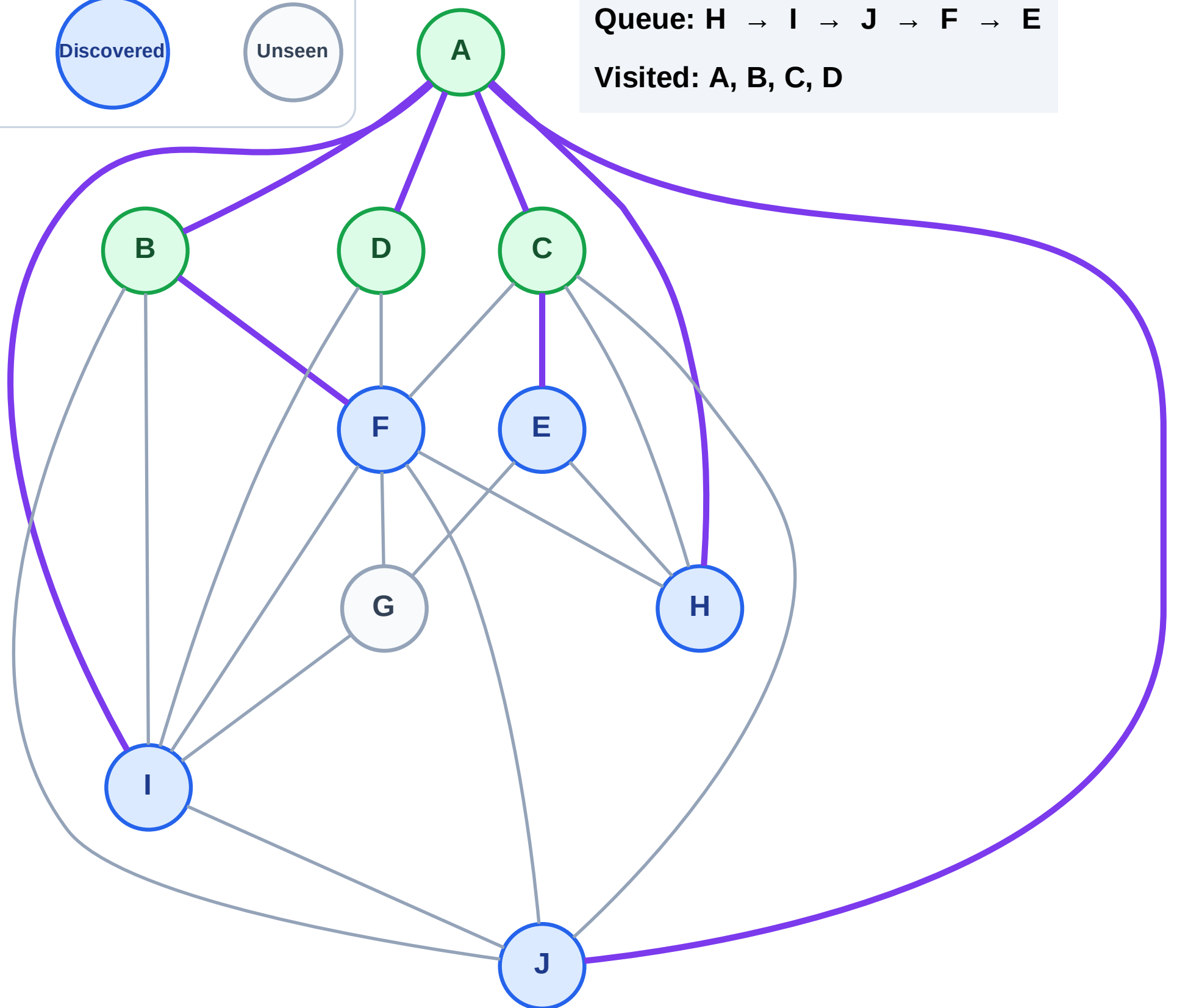
Step 9: No new neighbors from D

Legend



Queue: H → I → J → F → E

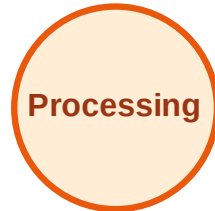
Visited: A, B, C, D



BFS Traversal

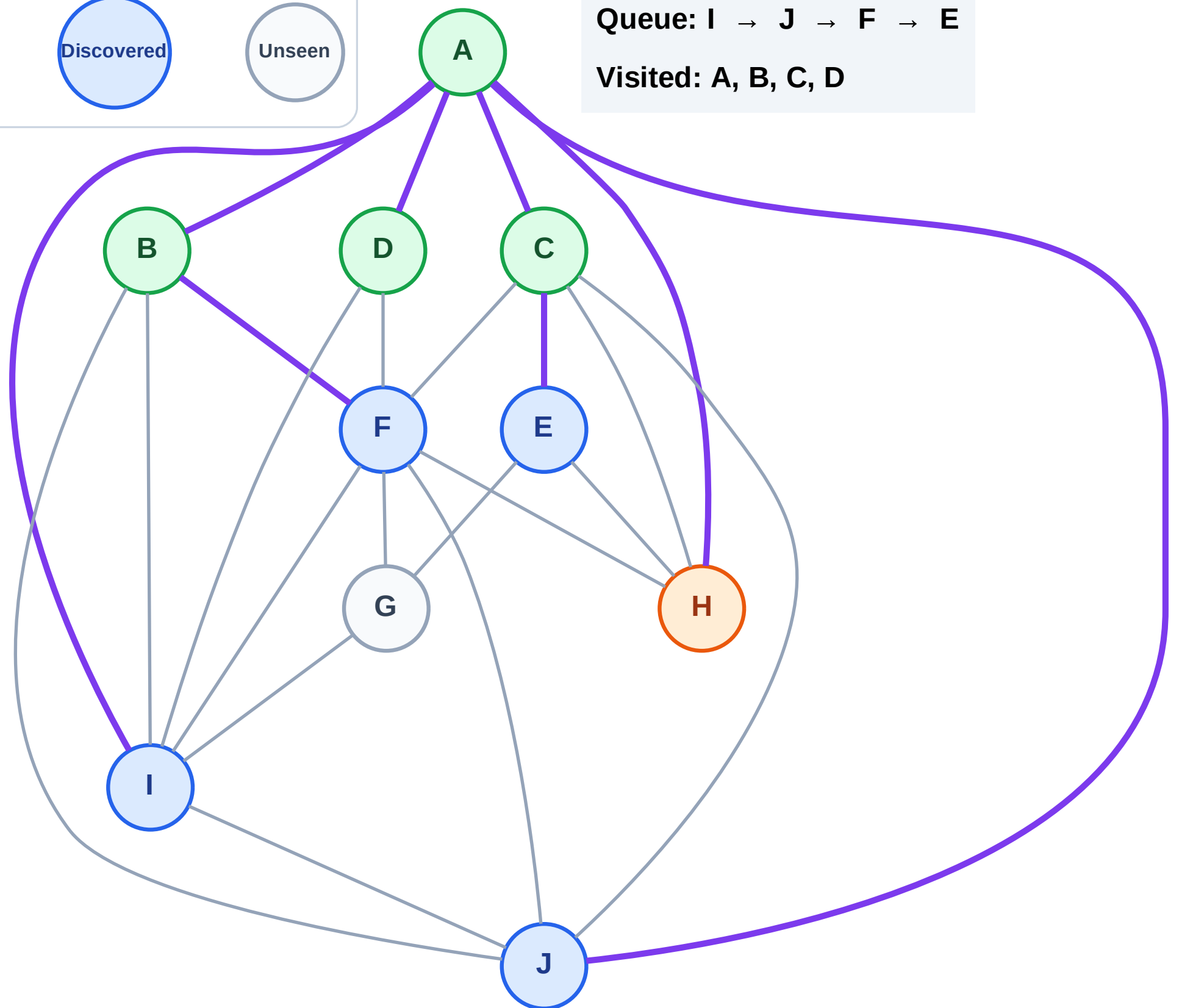
Step 10: Process node H

Legend



Queue: I → J → F → E

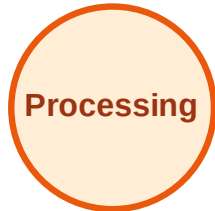
Visited: A, B, C, D



BFS Traversal

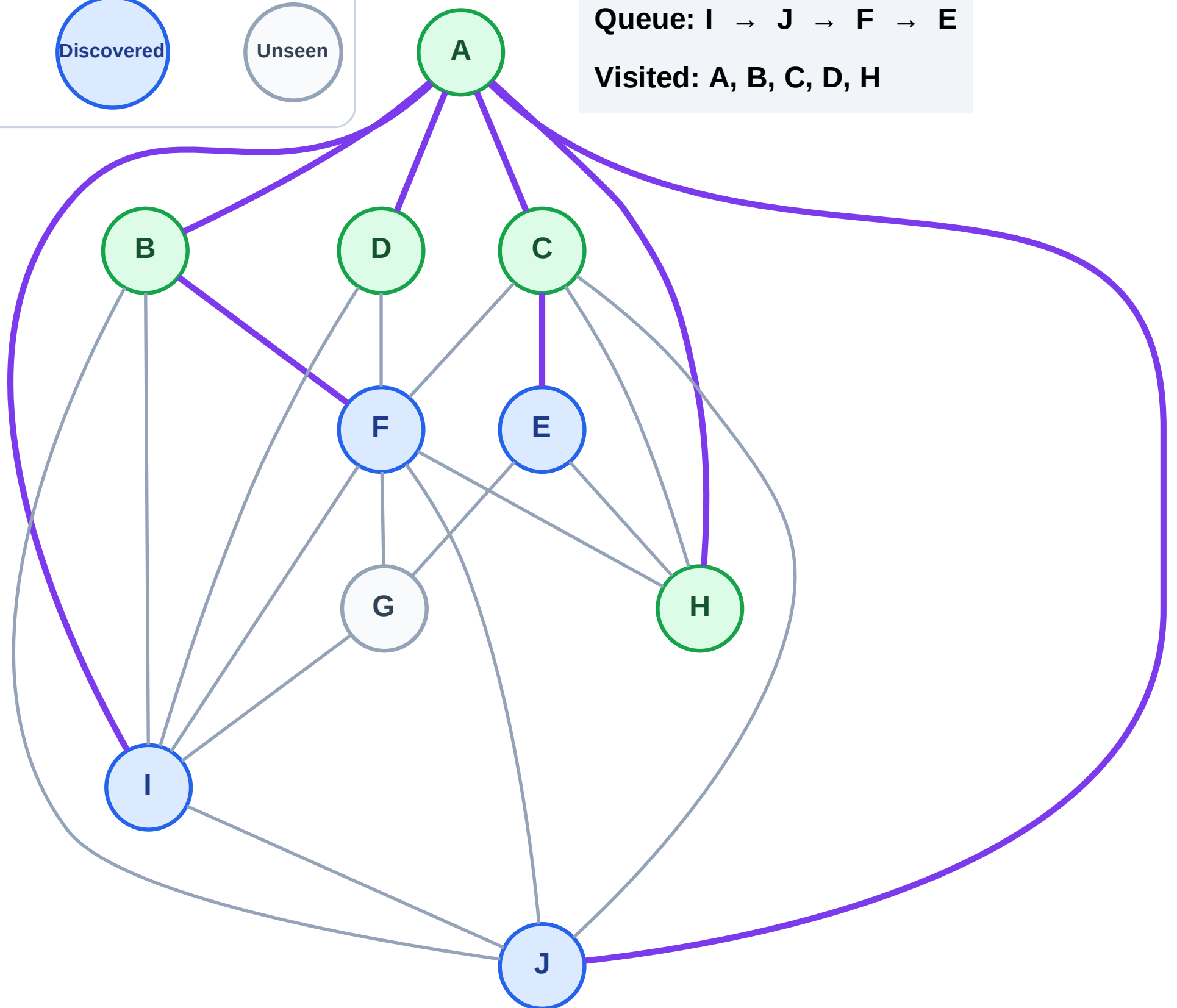
Step 11: No new neighbors from H

Legend



Queue: I → J → F → E

Visited: A, B, C, D, H



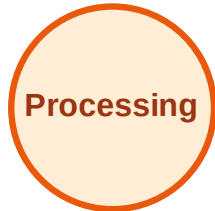
BFS Traversal

Step 12: Process node I

Legend



Complete



Processing



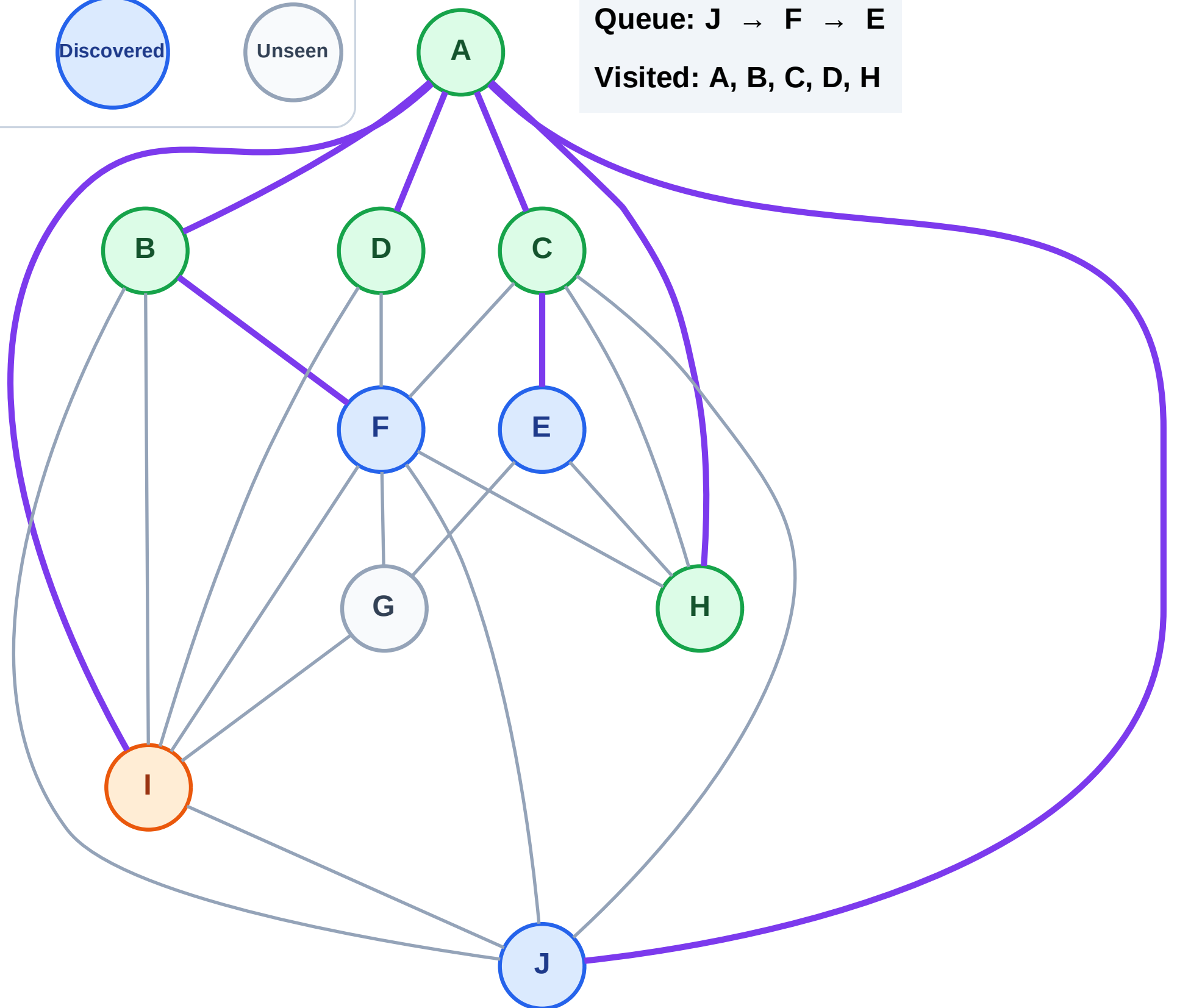
Discovered



Unseen

Queue: J → F → E

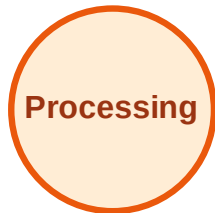
Visited: A, B, C, D, H



BFS Traversal

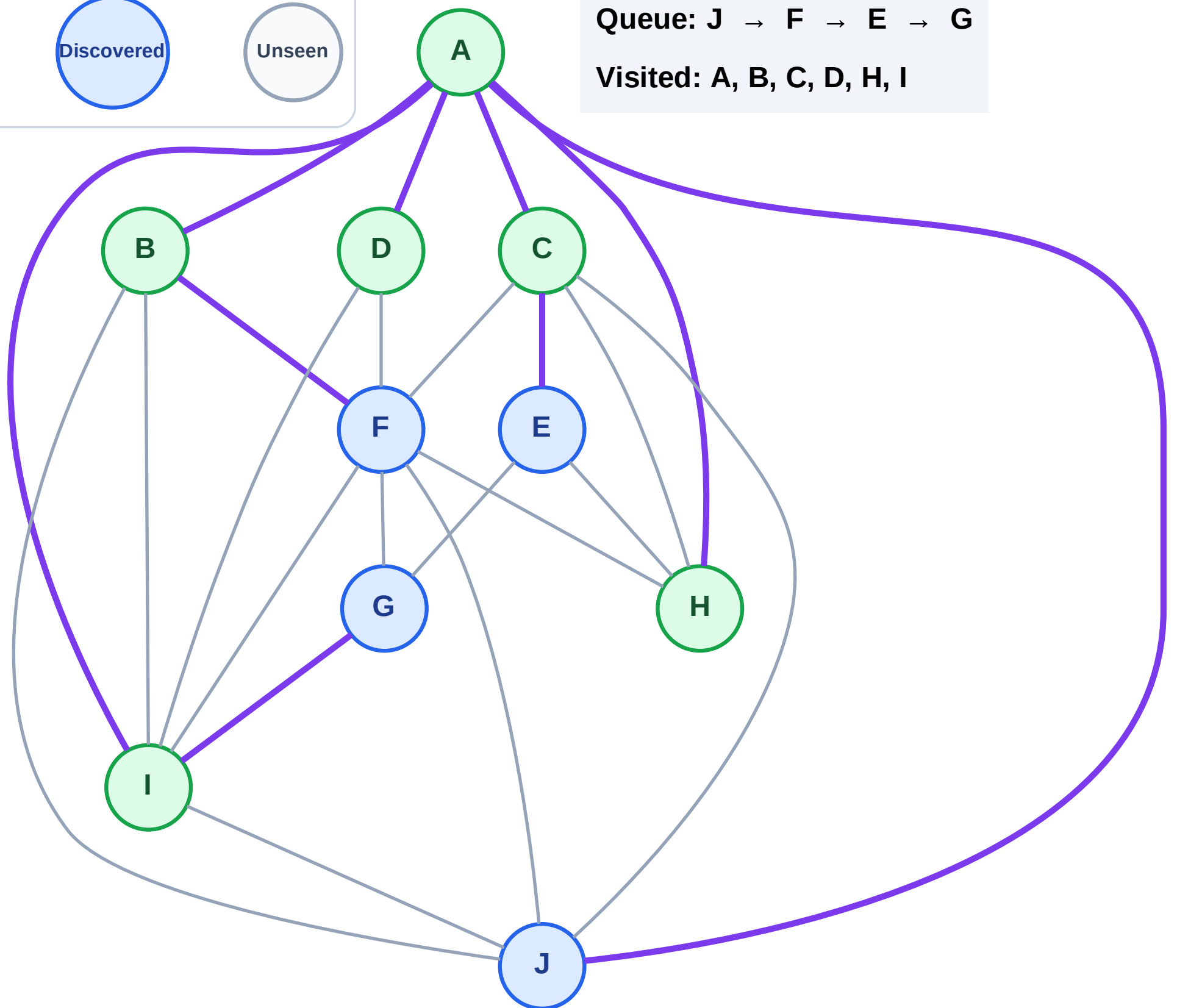
Step 13: Discover G from I

Legend



Queue: J → F → E → G

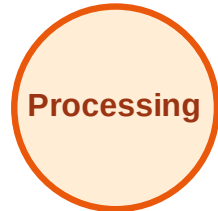
Visited: A, B, C, D, H, I



BFS Traversal

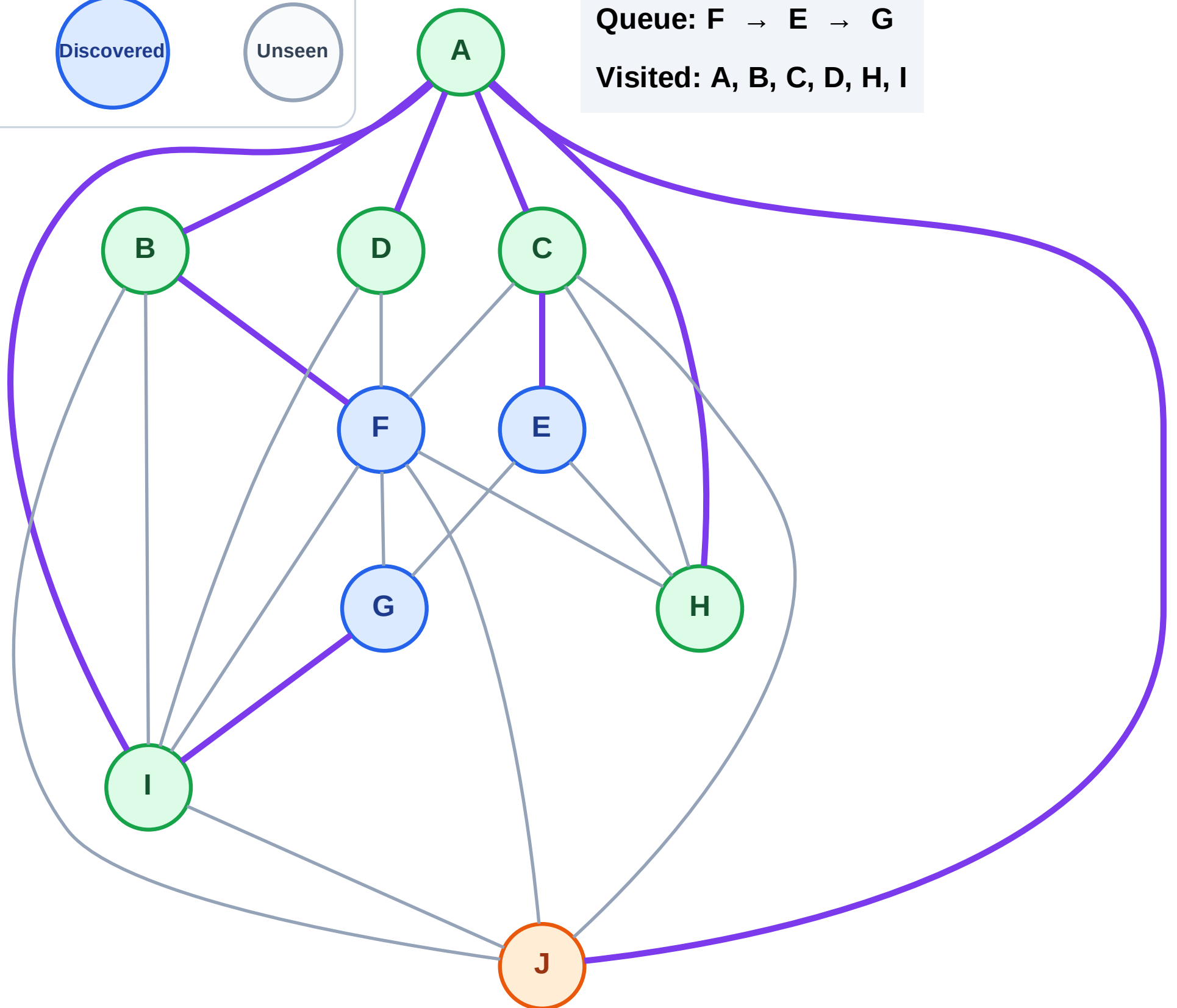
Step 14: Process node J

Legend



Queue: F → E → G

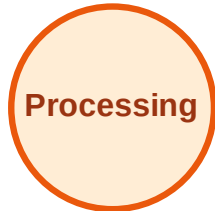
Visited: A, B, C, D, H, I



BFS Traversal

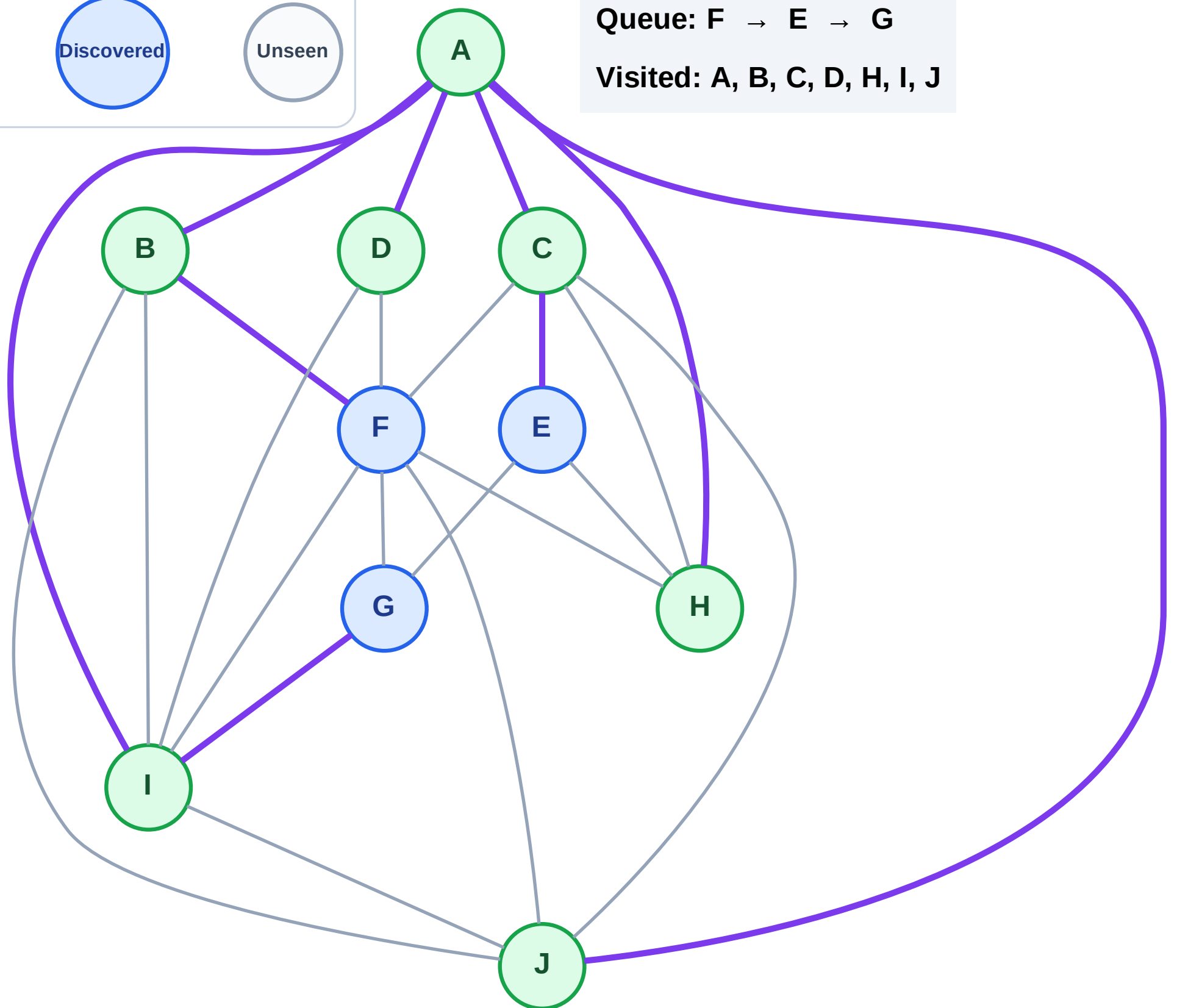
Step 15: No new neighbors from J

Legend



Queue: F → E → G

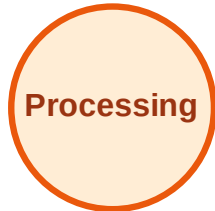
Visited: A, B, C, D, H, I, J



BFS Traversal

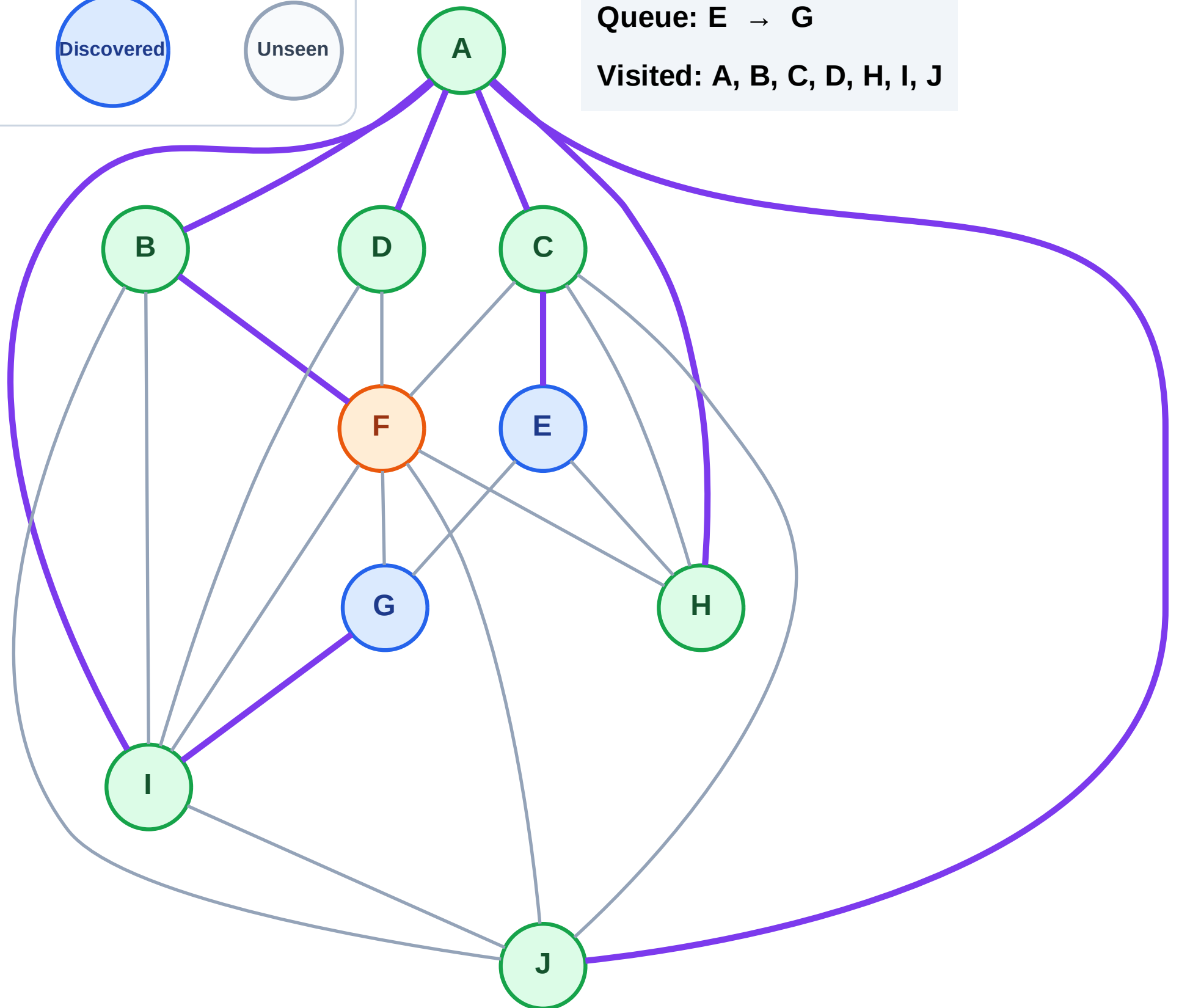
Step 16: Process node F

Legend



Queue: E → G

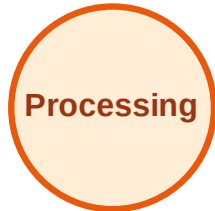
Visited: A, B, C, D, H, I, J



BFS Traversal

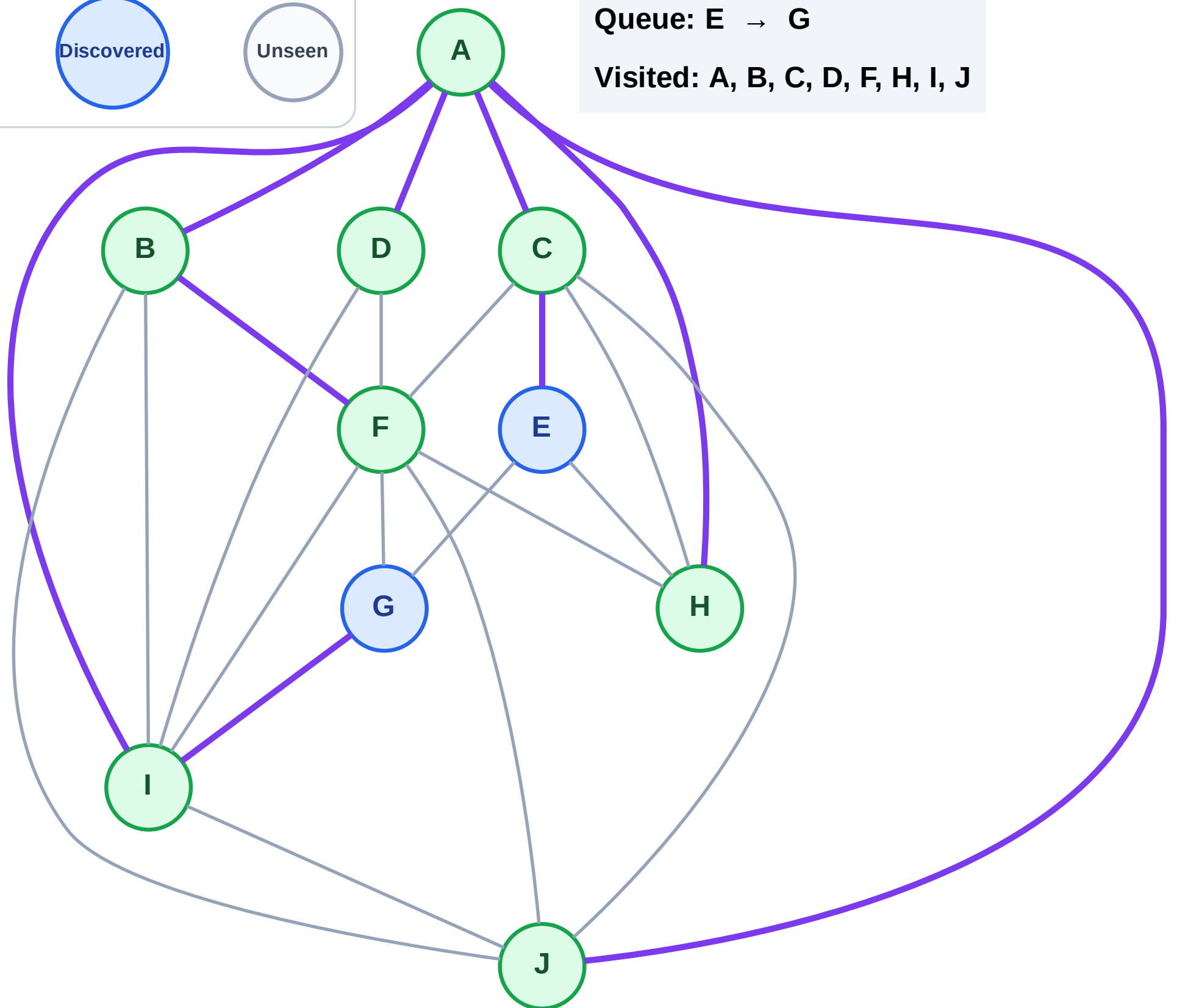
Step 17: No new neighbors from F

Legend



Queue: E → G

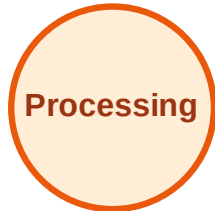
Visited: A, B, C, D, F, H, I, J



BFS Traversal

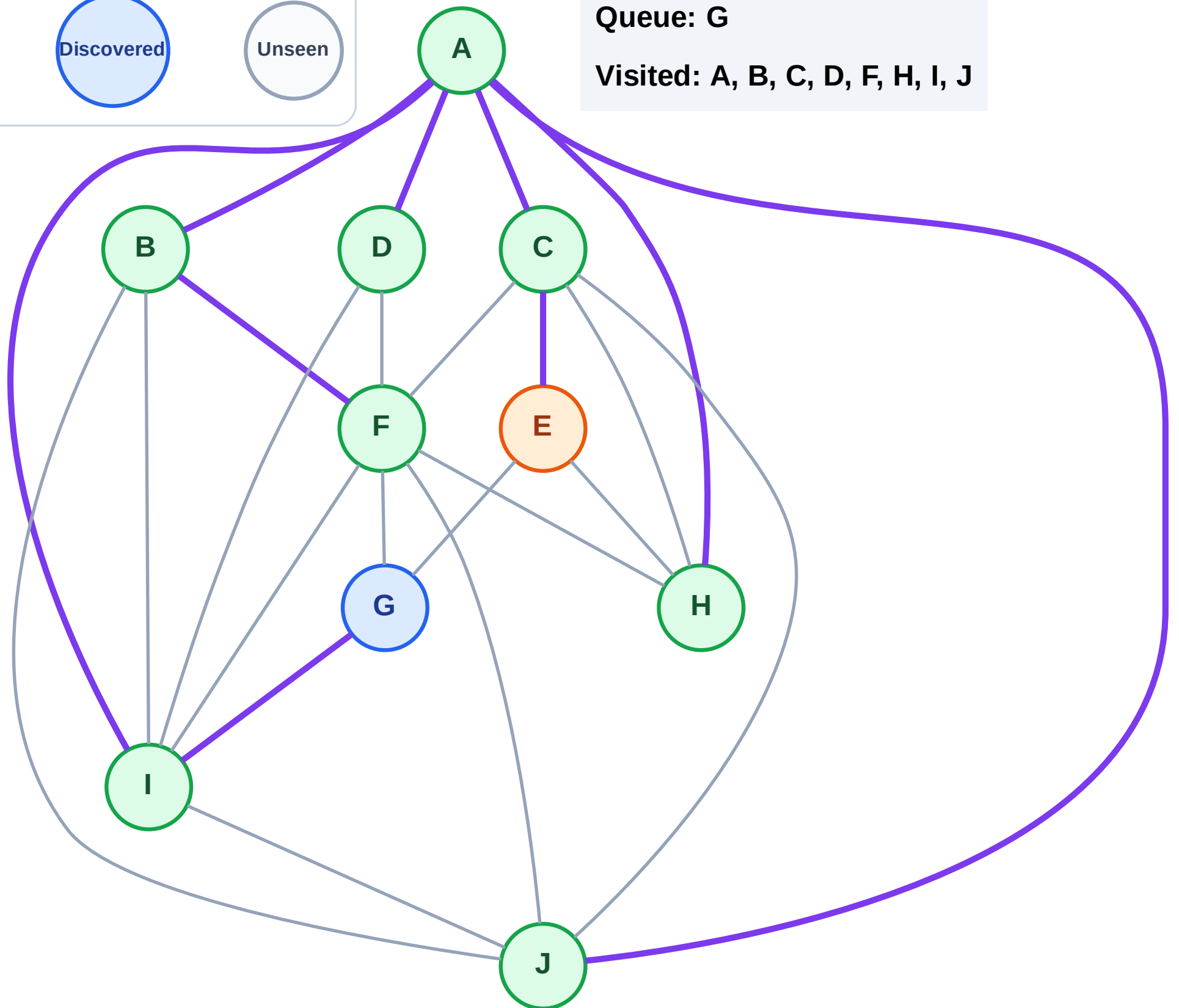
Step 18: Process node E

Legend



Queue: G

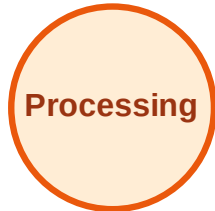
Visited: A, B, C, D, F, H, I, J



BFS Traversal

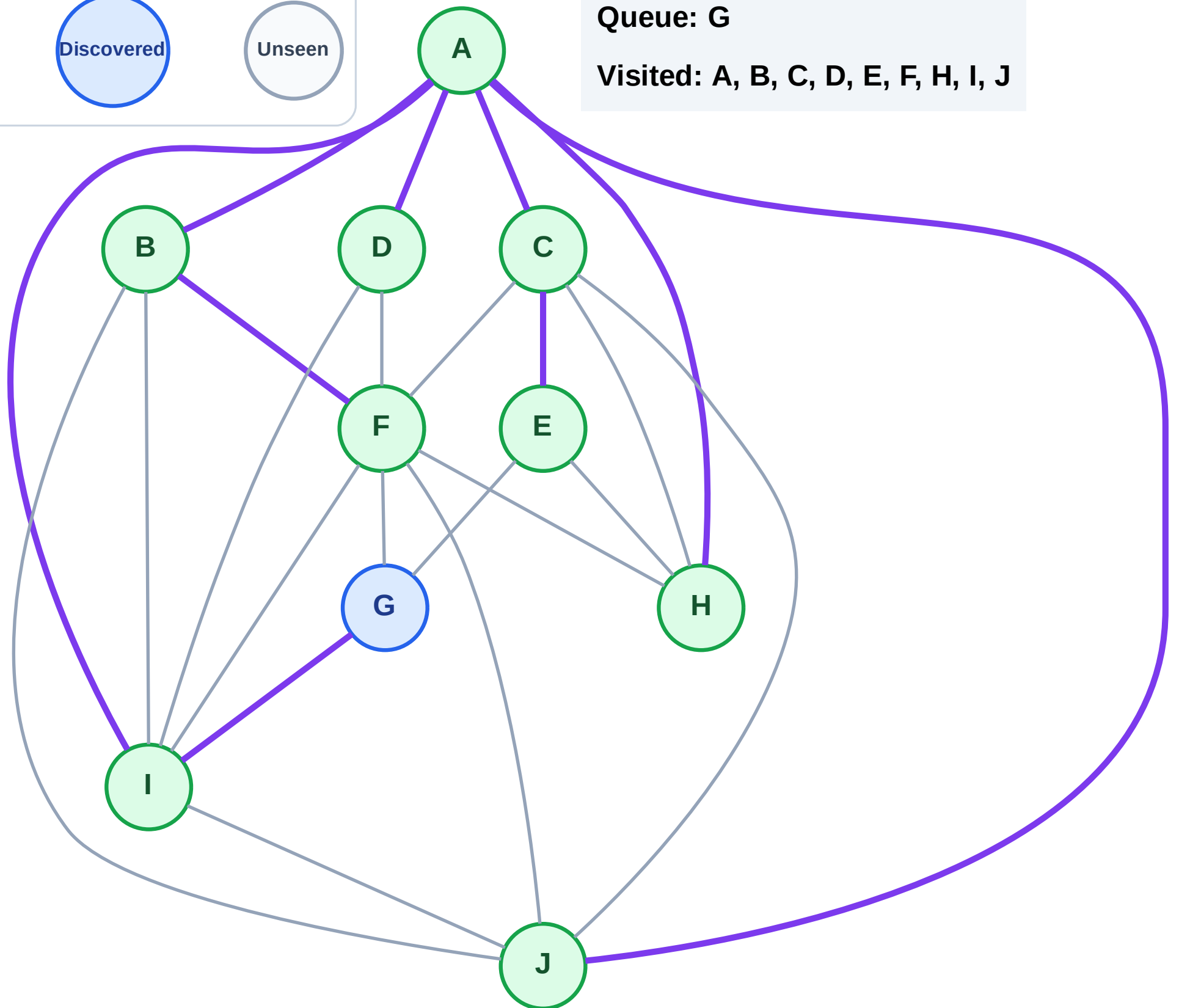
Step 19: No new neighbors from E

Legend



Queue: G

Visited: A, B, C, D, E, F, H, I, J

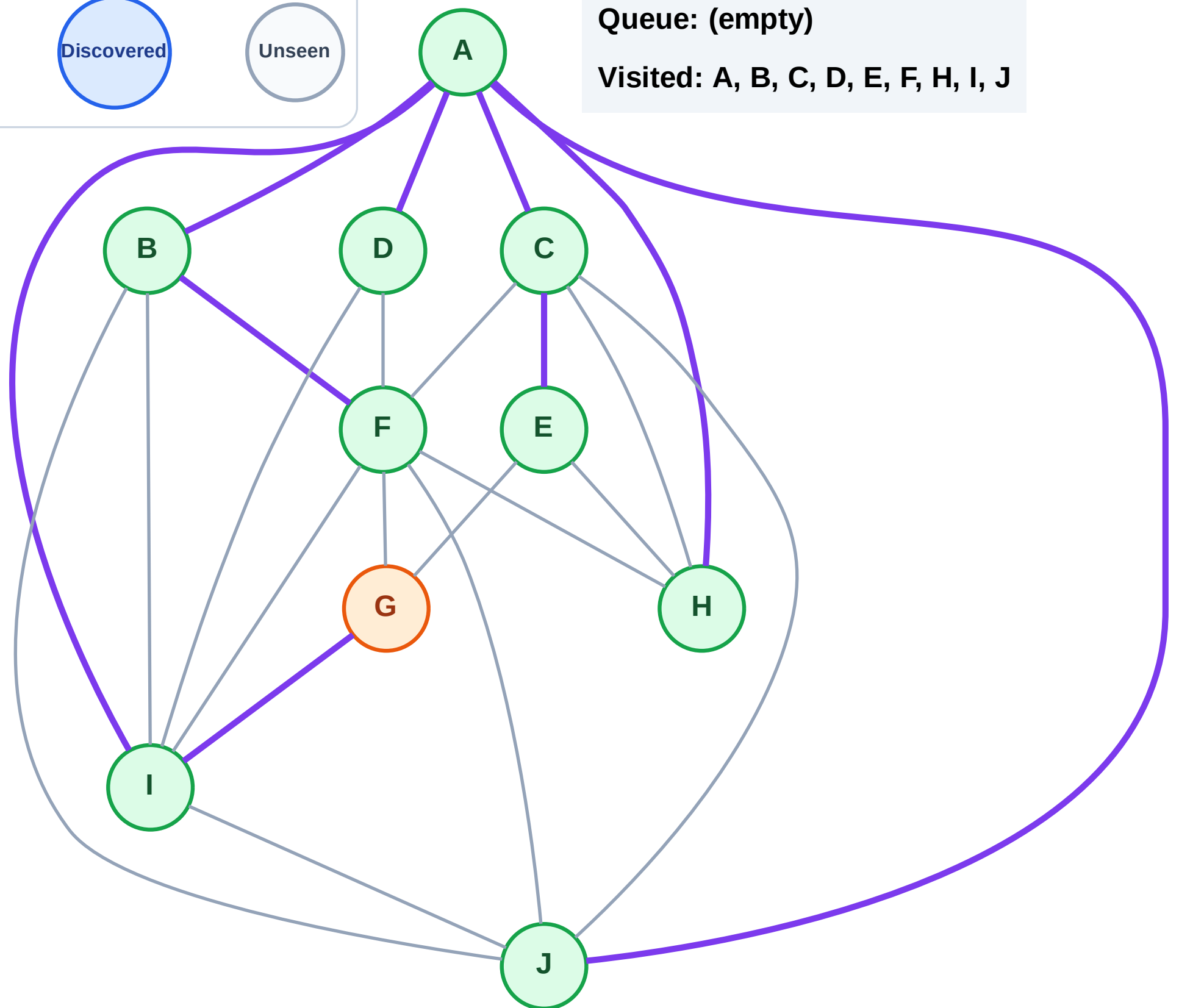


Step 20: Process node G

Legend

Unseen

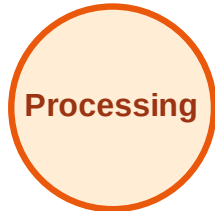
Visited: A, B, C, D, E, F, H, I, J



BFS Traversal

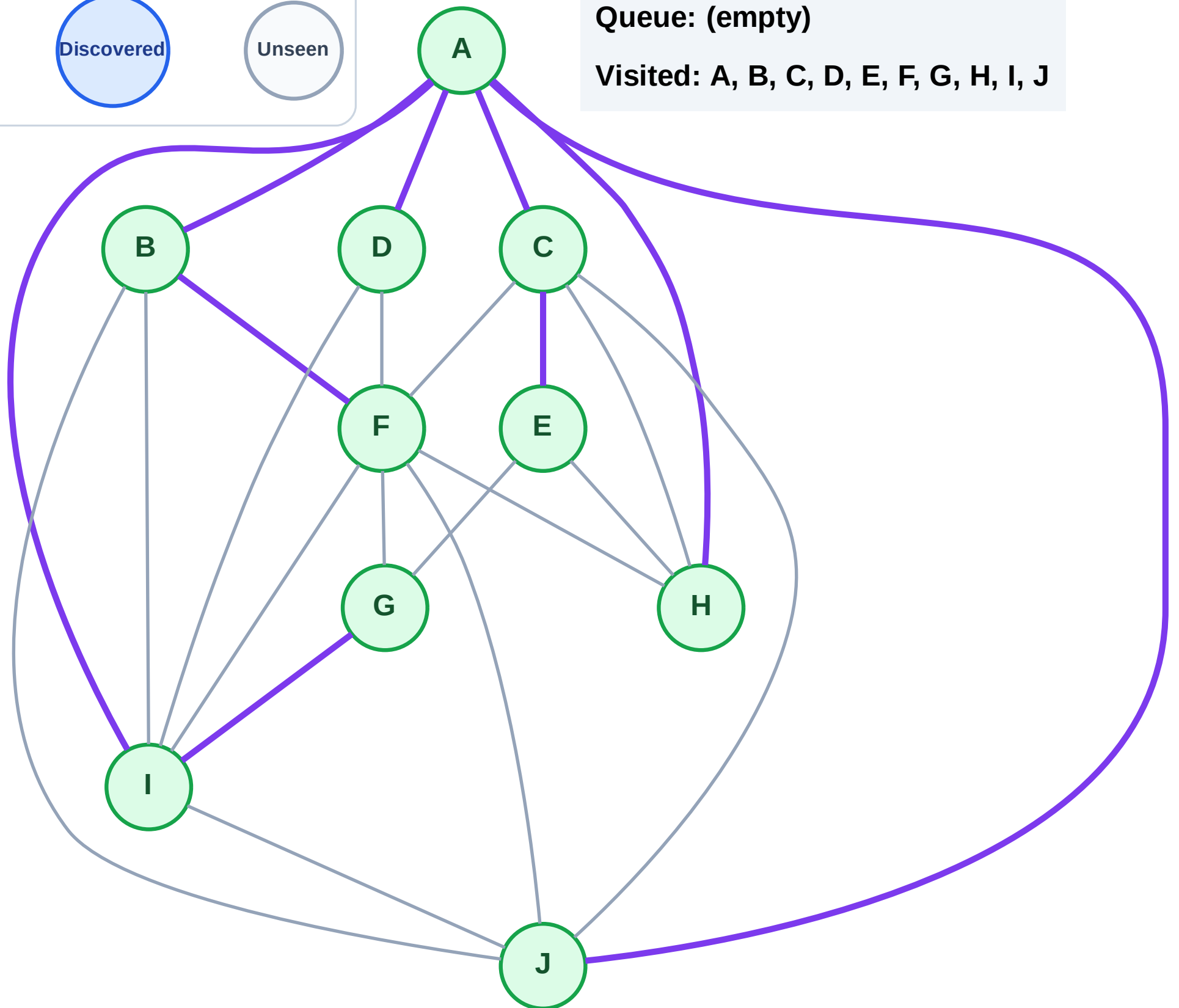
Step 21: No new neighbors from G

Legend



Queue: (empty)

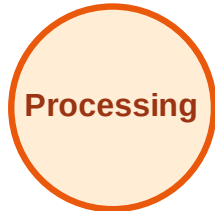
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

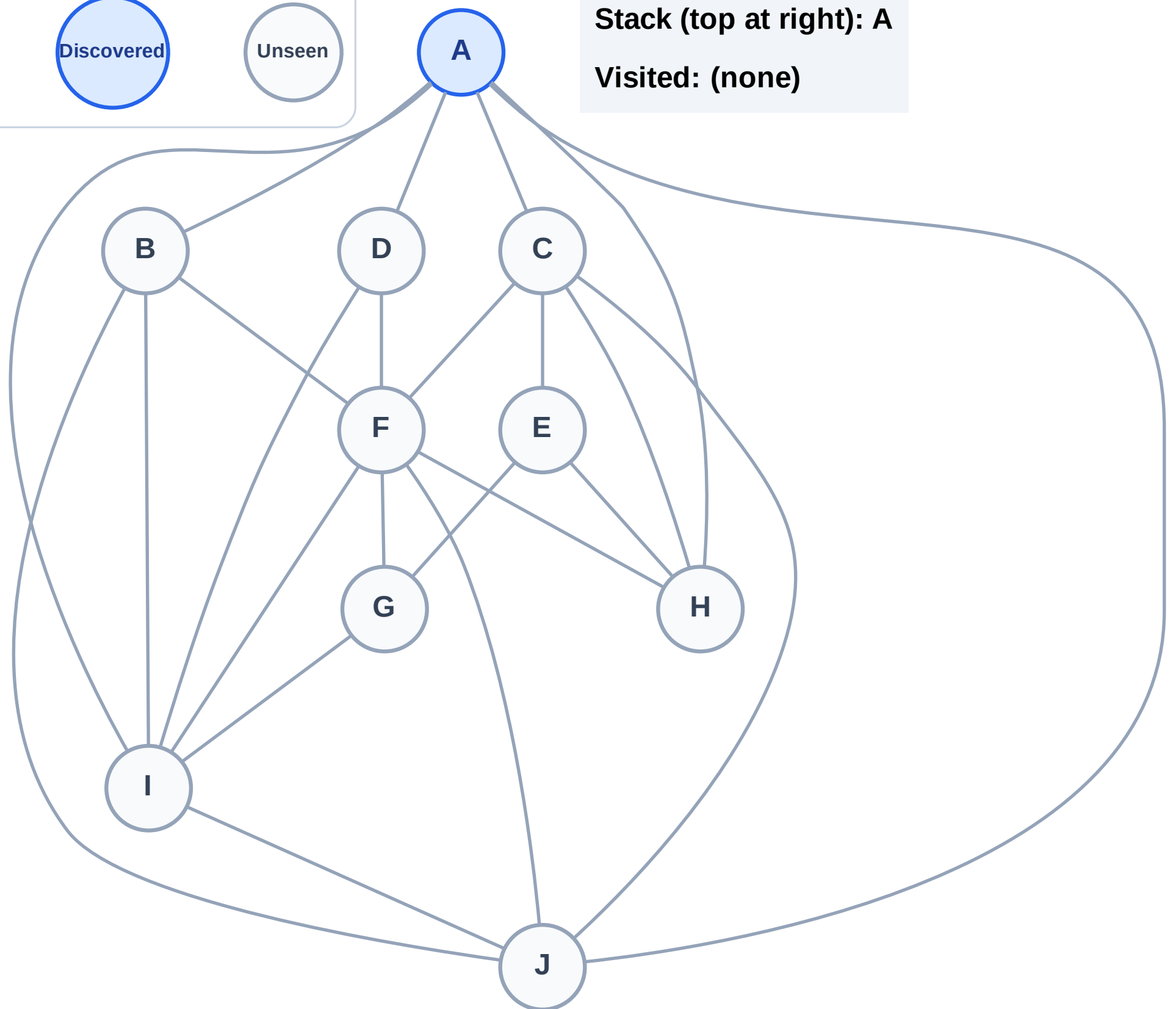
Step 1: Start at A

Legend



Stack (top at right): A

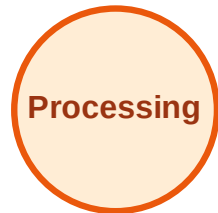
Visited: (none)



DFS Traversal

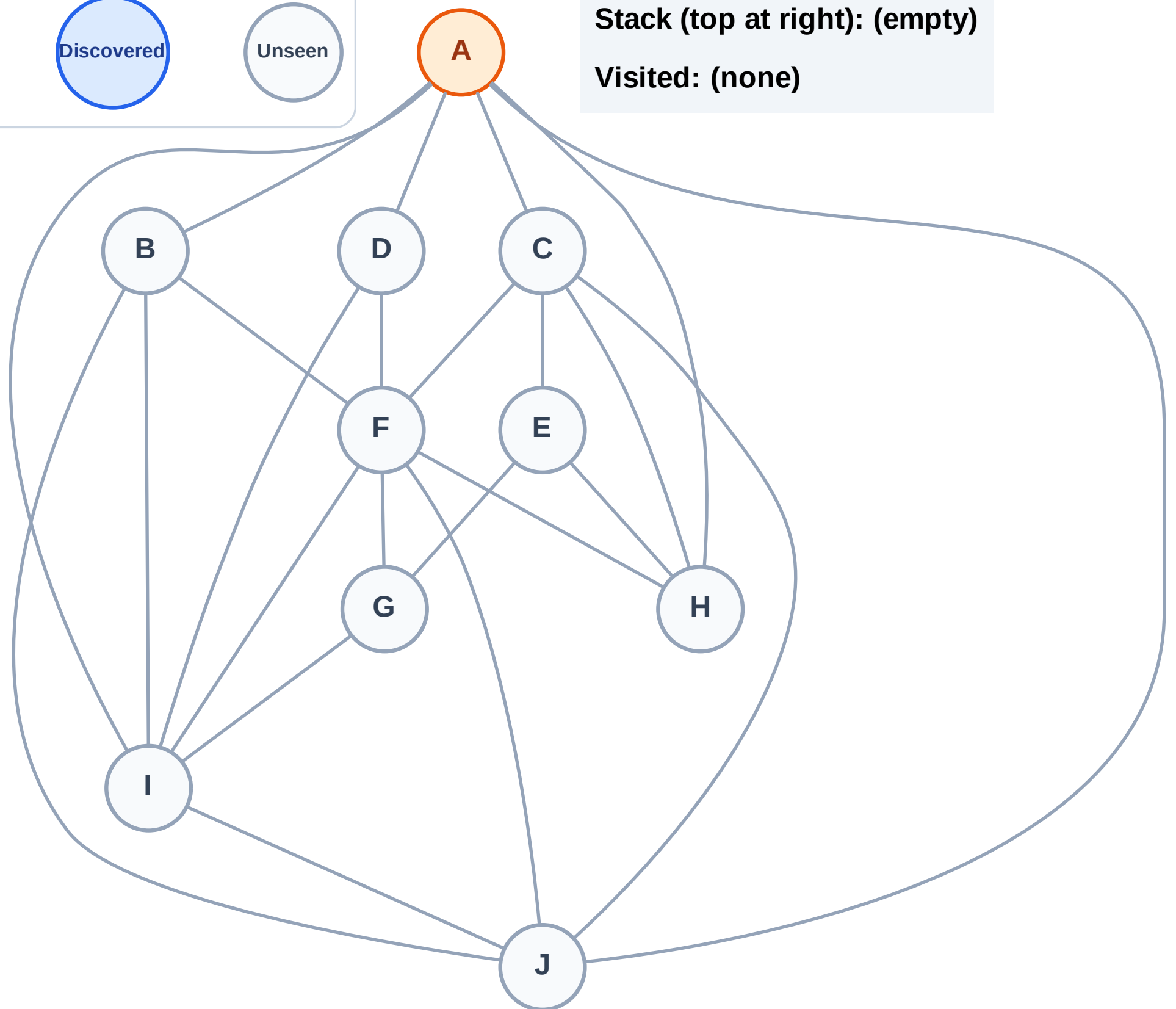
Step 2: Process node A

Legend



Stack (top at right): (empty)

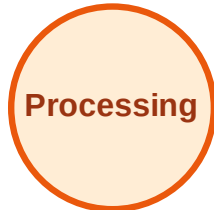
Visited: (none)



DFS Traversal

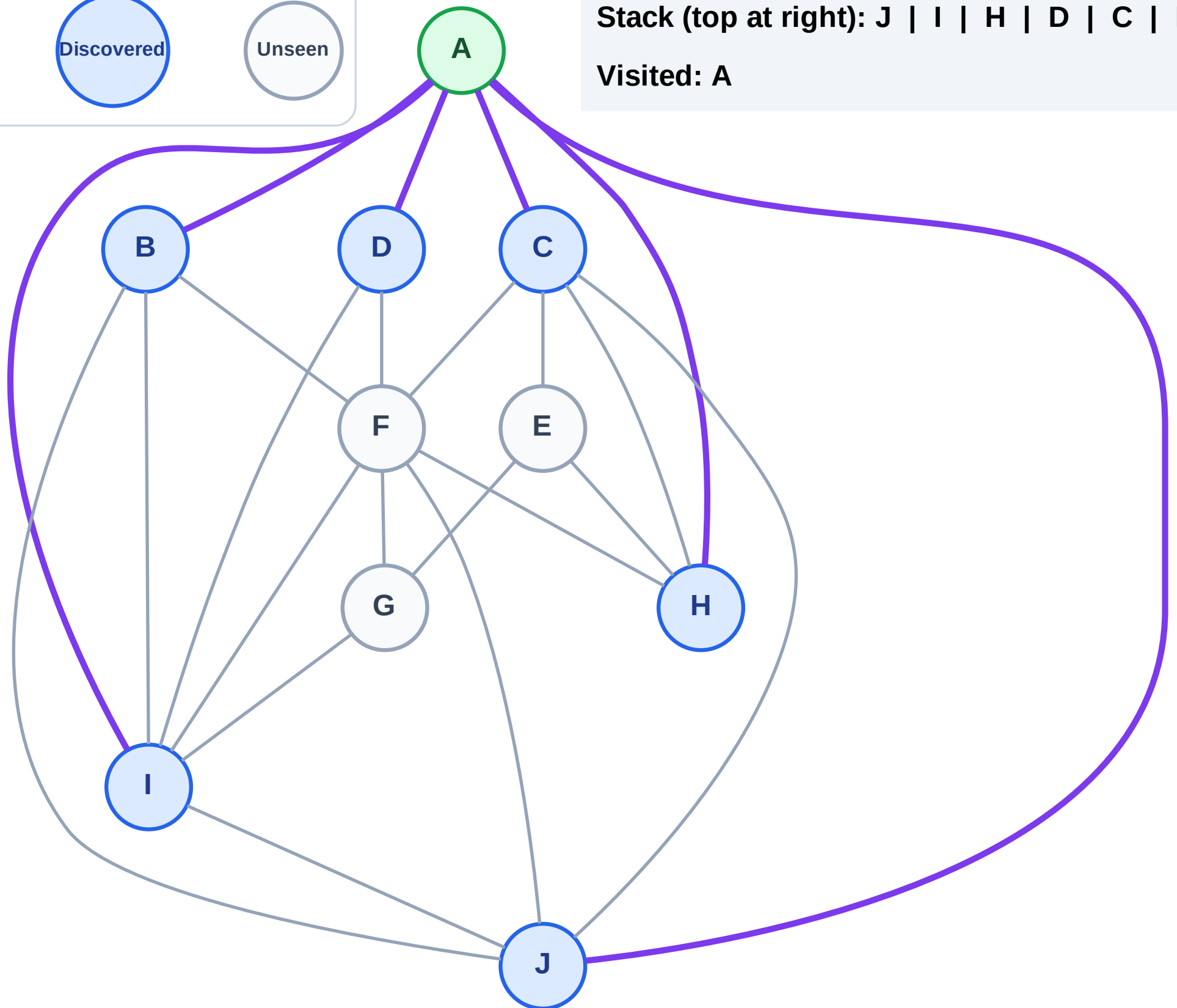
Step 3: Discover J, I, H, D, C, B from A

Legend



Stack (top at right): J | I | H | D | C | B

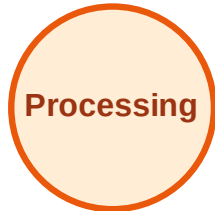
Visited: A



DFS Traversal

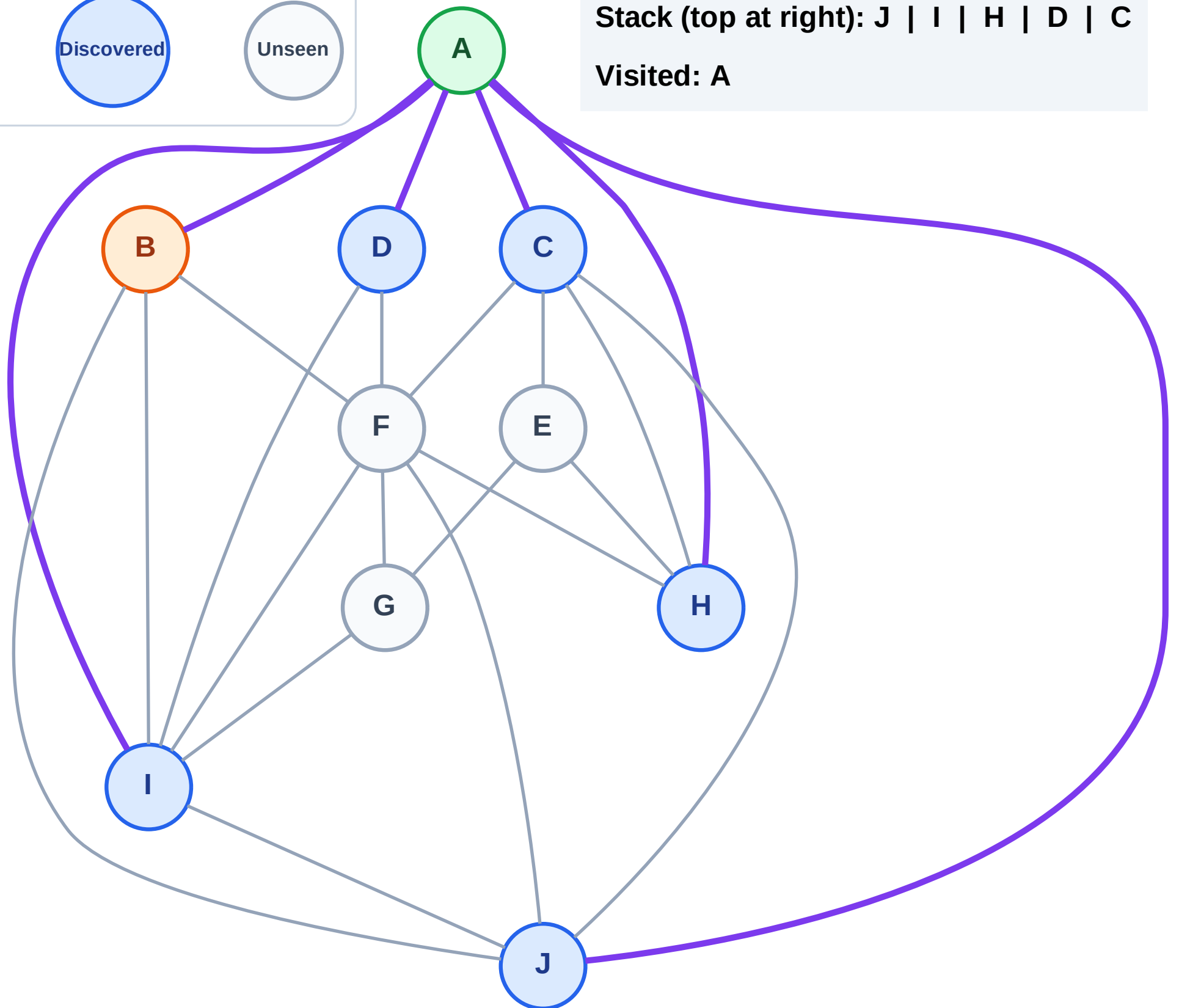
Step 4: Process node B

Legend



Stack (top at right): J | I | H | D | C

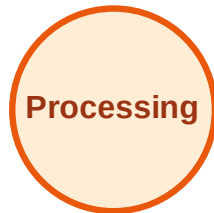
Visited: A



DFS Traversal

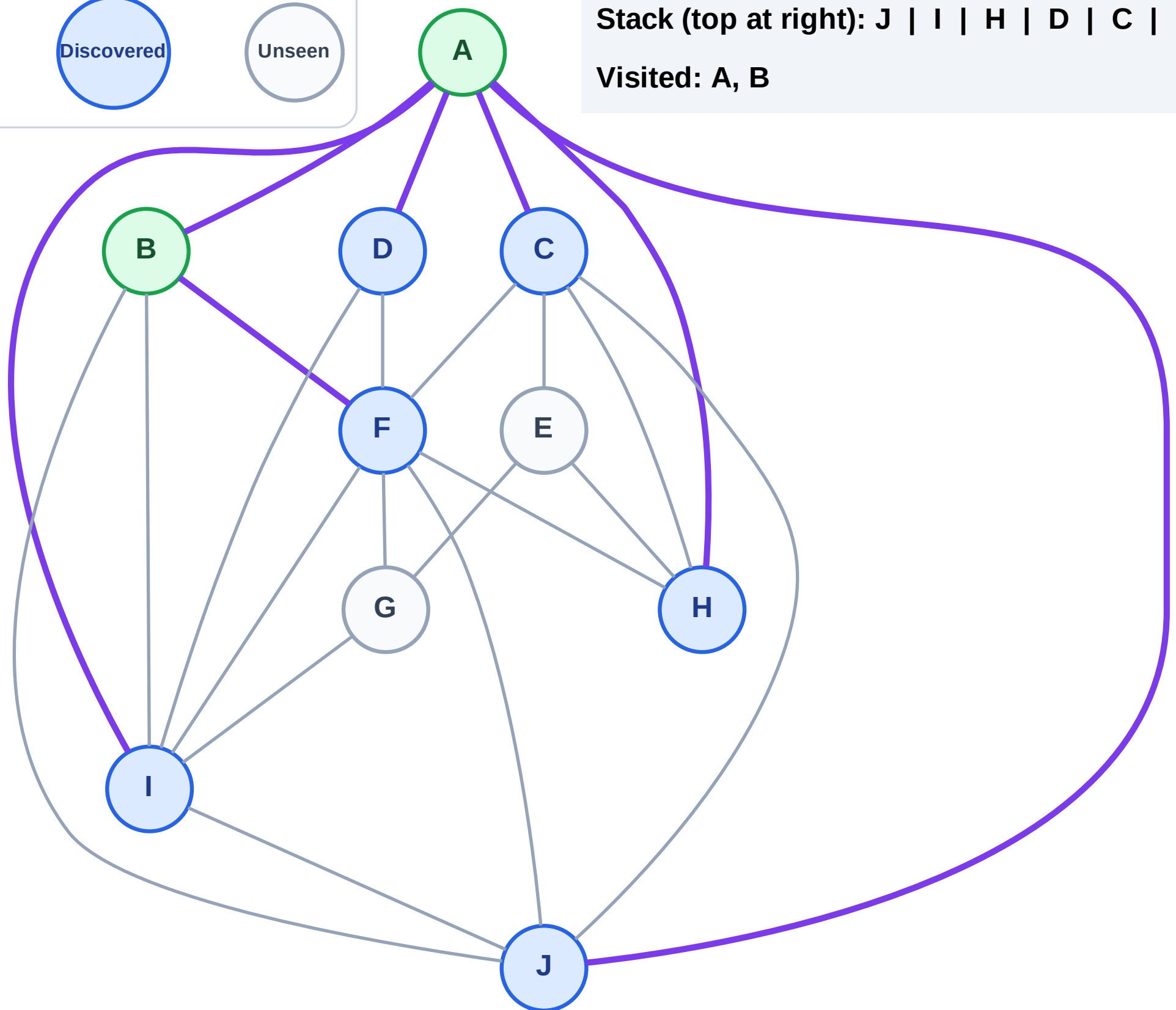
Step 5: Discover F from B

Legend



Stack (top at right): J | I | H | D | C | F

Visited: A, B



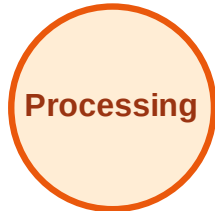
DFS Traversal

Step 6: Process node F

Legend



Complete



Processing



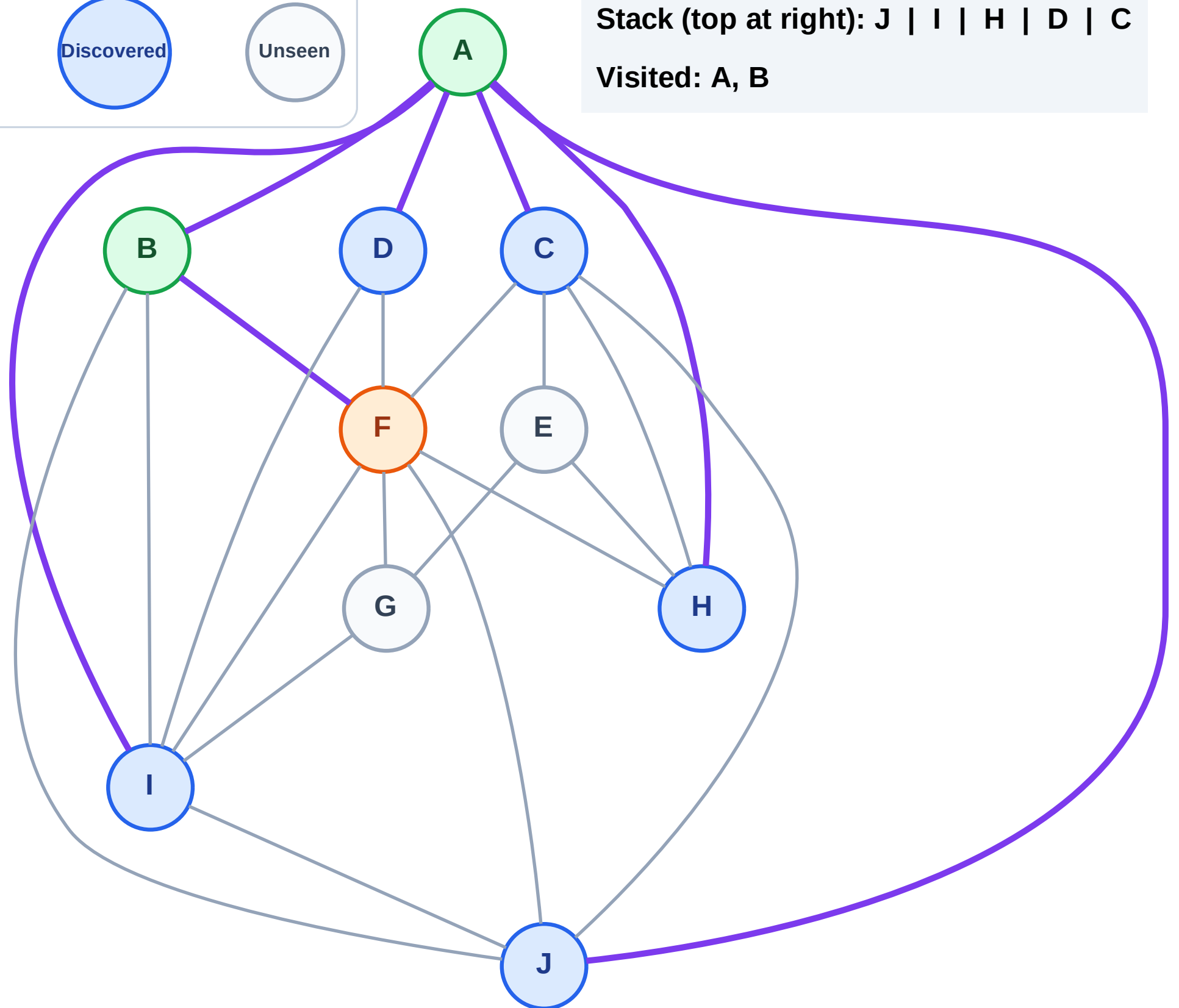
Discovered



Unseen

Stack (top at right): J | I | H | D | C

Visited: A, B



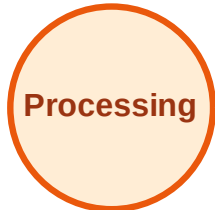
DFS Traversal

Step 7: Discover G from F

Legend



Complete



Processing



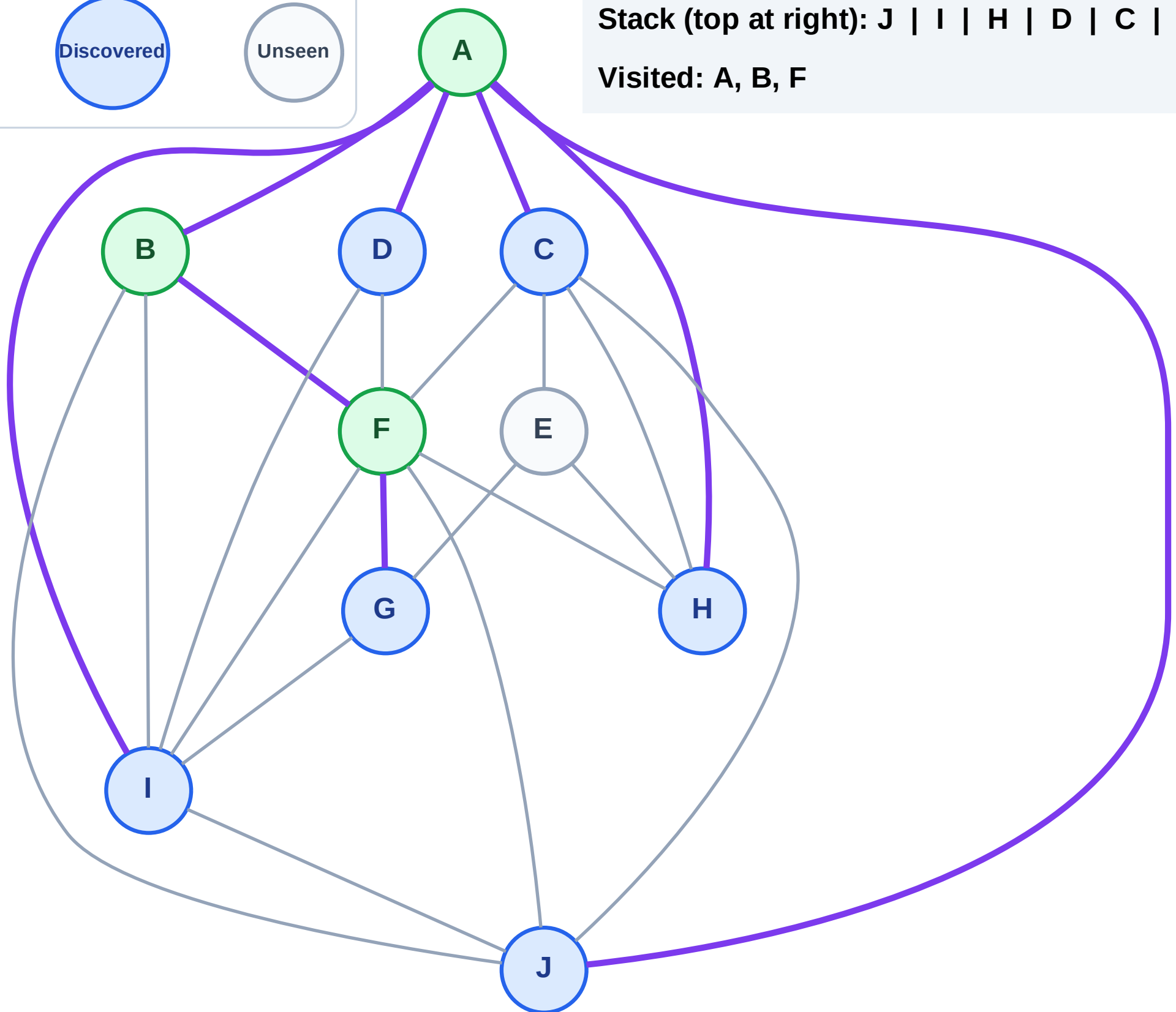
Discovered



Unseen

Stack (top at right): J | I | H | D | C | G

Visited: A, B, F



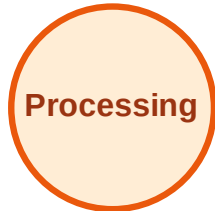
DFS Traversal

Step 8: Process node G

Legend



Complete



Processing



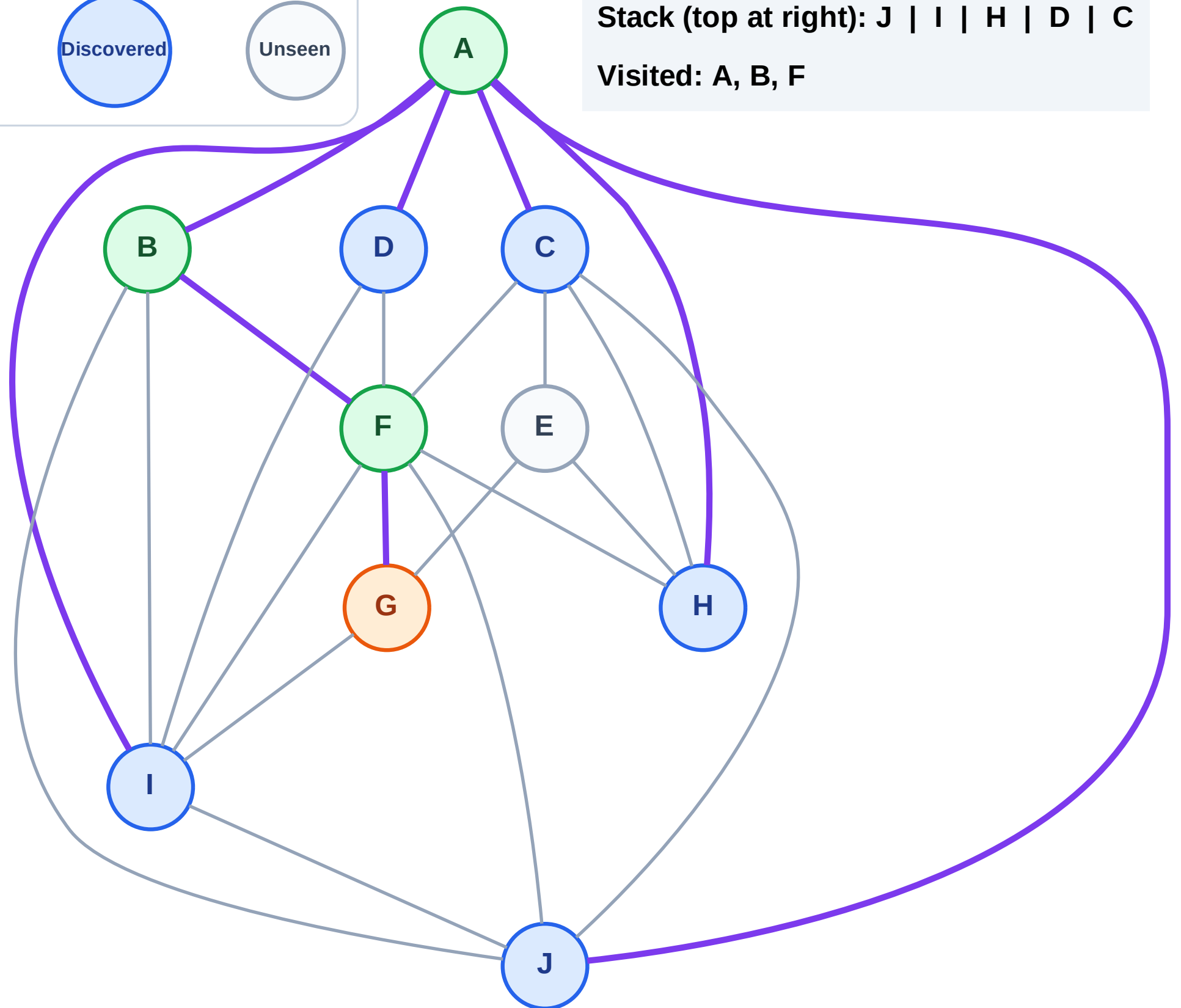
Discovered



Unseen

Stack (top at right): J | I | H | D | C

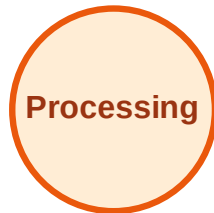
Visited: A, B, F



DFS Traversal

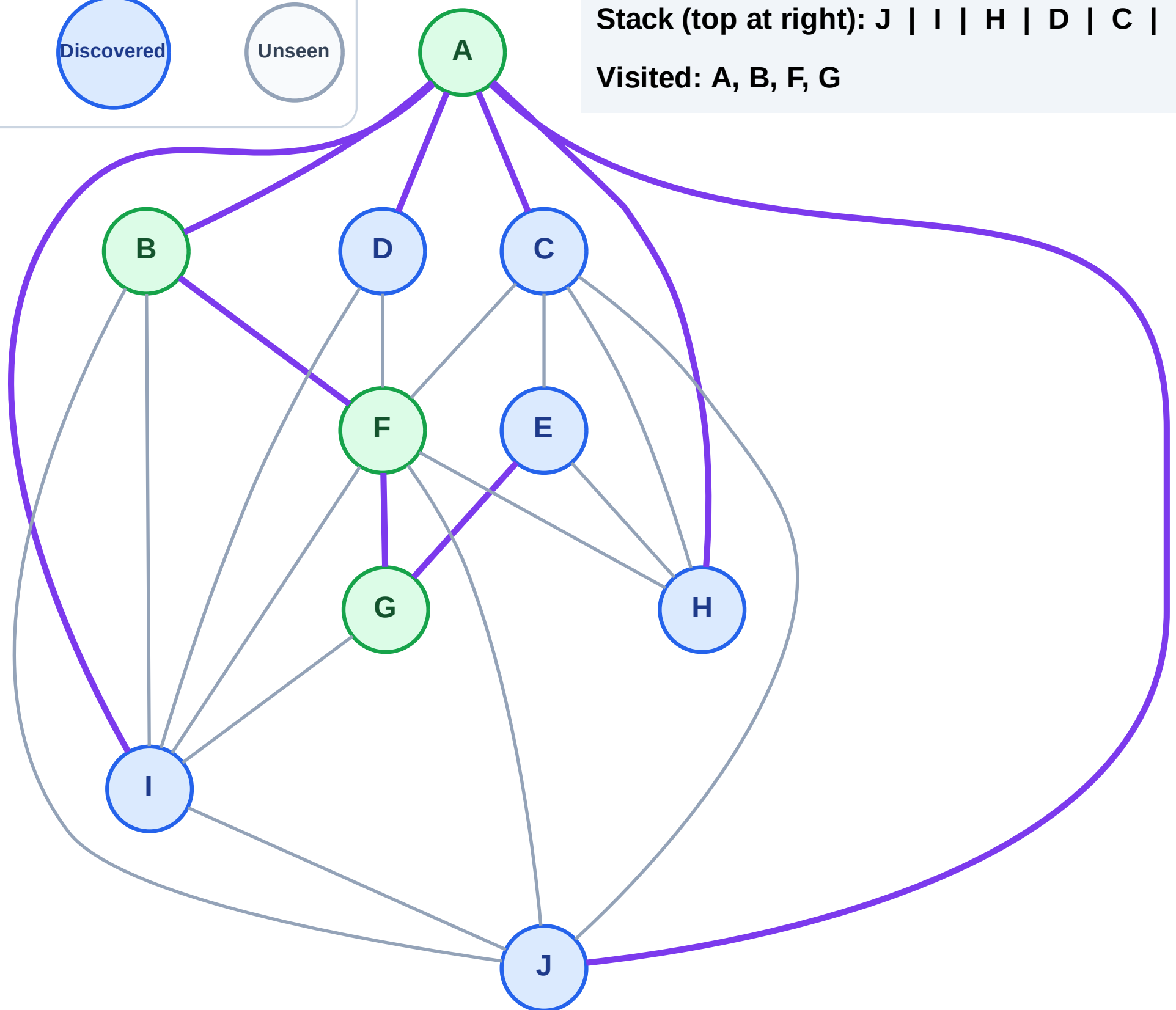
Step 9: Discover E from G

Legend



Stack (top at right): J | I | H | D | C | E

Visited: A, B, F, G



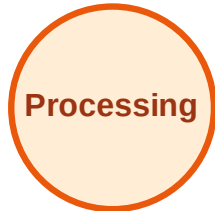
DFS Traversal

Step 10: Process node E

Legend



Complete



Processing



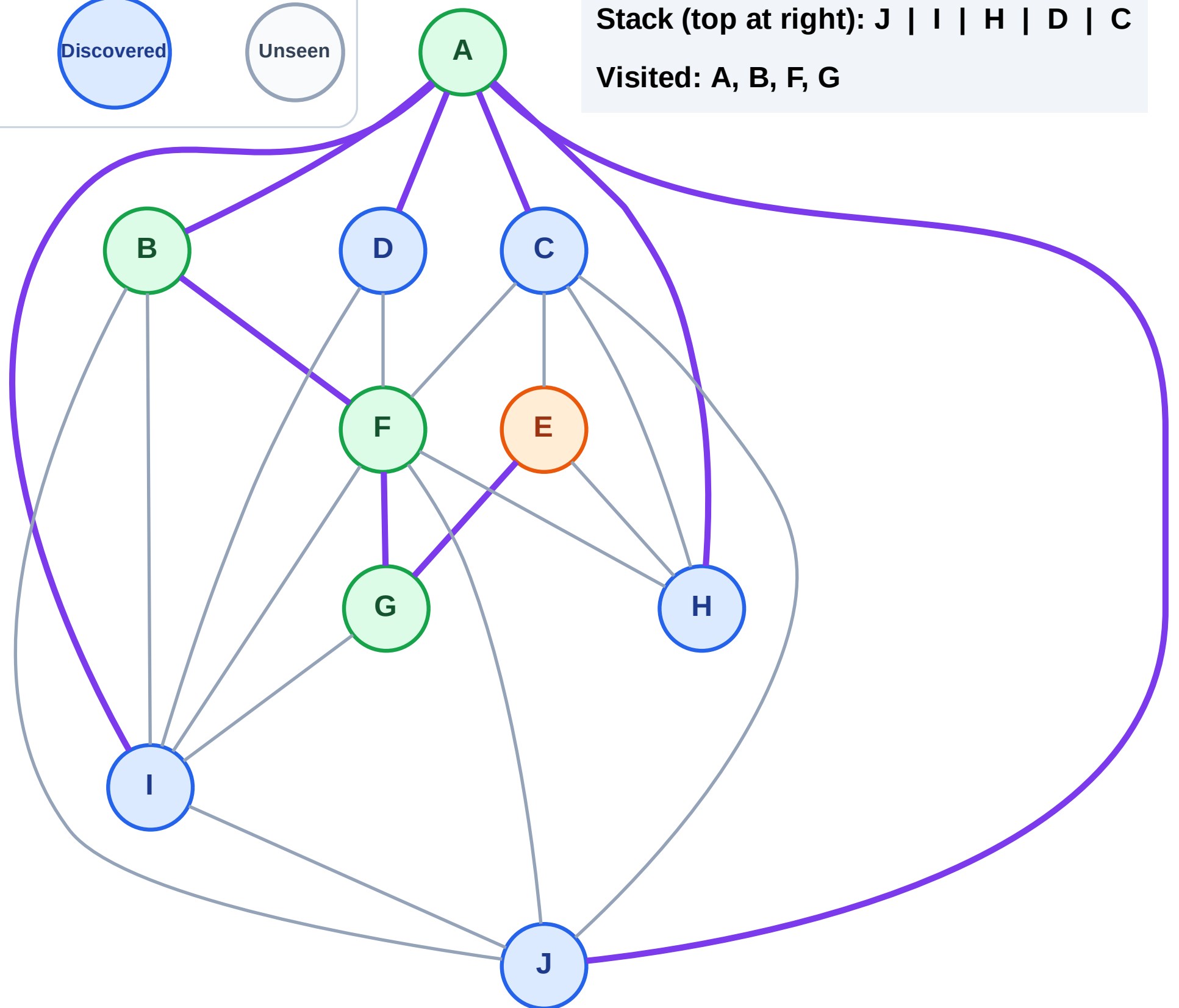
Discovered



Unseen

Stack (top at right): J | I | H | D | C

Visited: A, B, F, G



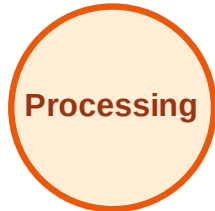
DFS Traversal

Step 11: No new neighbors from E

Legend



Complete



Processing



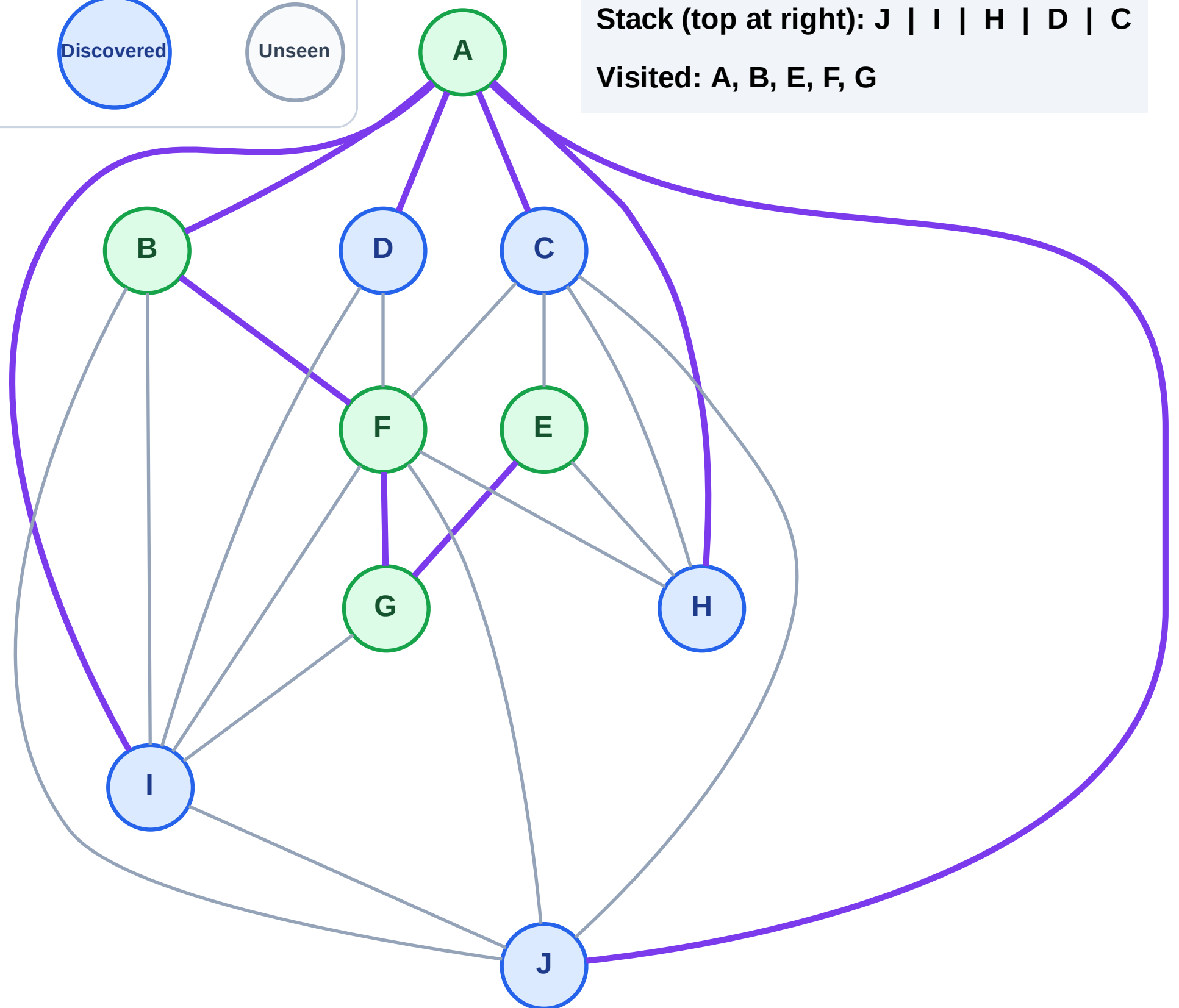
Discovered



Unseen

Stack (top at right): J | I | H | D | C

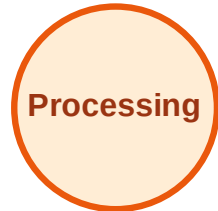
Visited: A, B, E, F, G



DFS Traversal

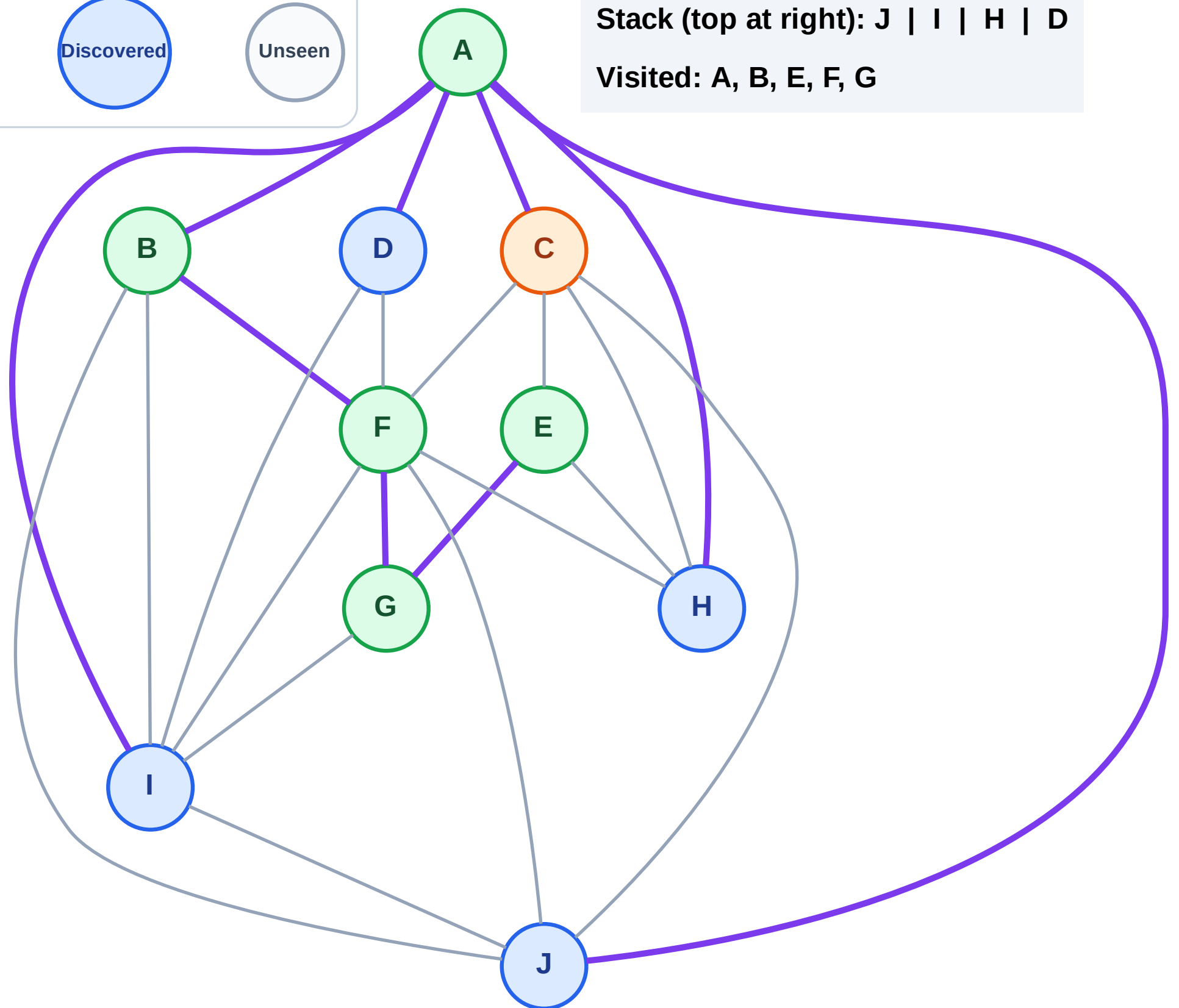
Step 12: Process node C

Legend



Stack (top at right): J | I | H | D

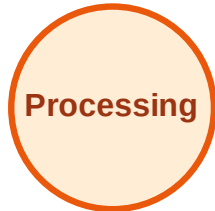
Visited: A, B, E, F, G



DFS Traversal

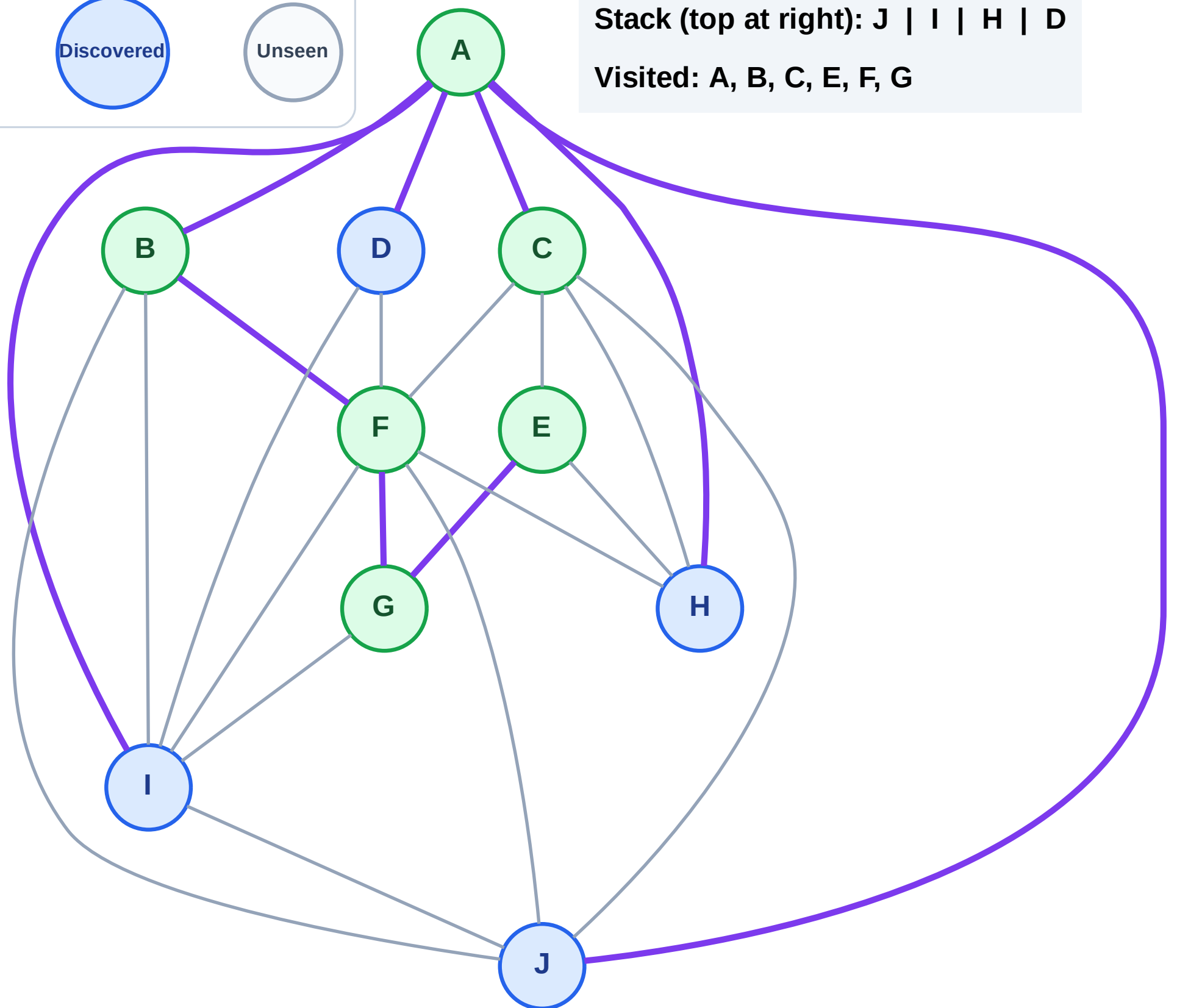
Step 13: No new neighbors from C

Legend



Stack (top at right): J | I | H | D

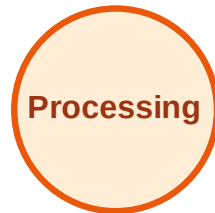
Visited: A, B, C, E, F, G



DFS Traversal

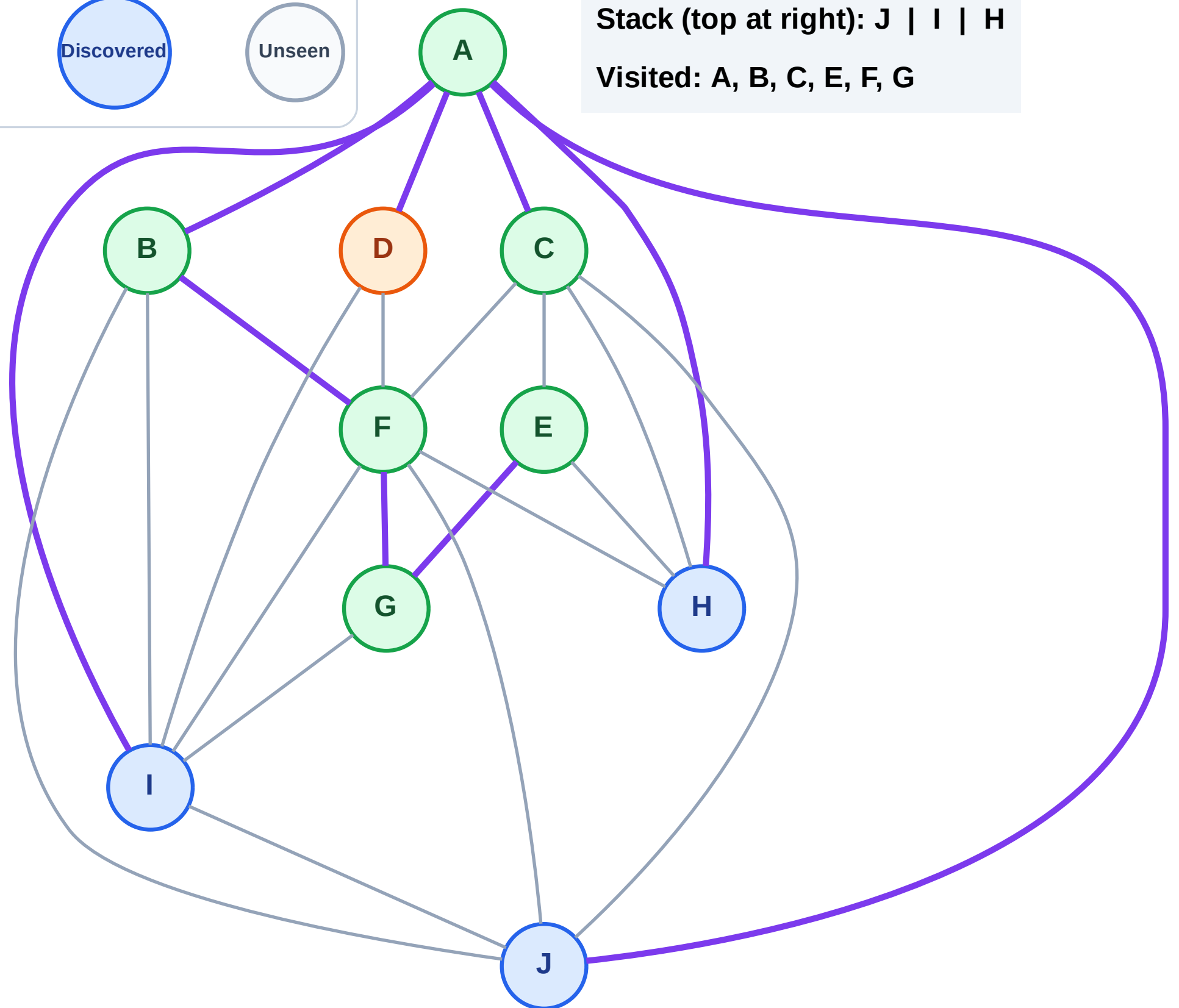
Step 14: Process node D

Legend



Stack (top at right): J | I | H

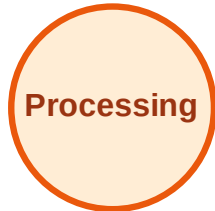
Visited: A, B, C, E, F, G



DFS Traversal

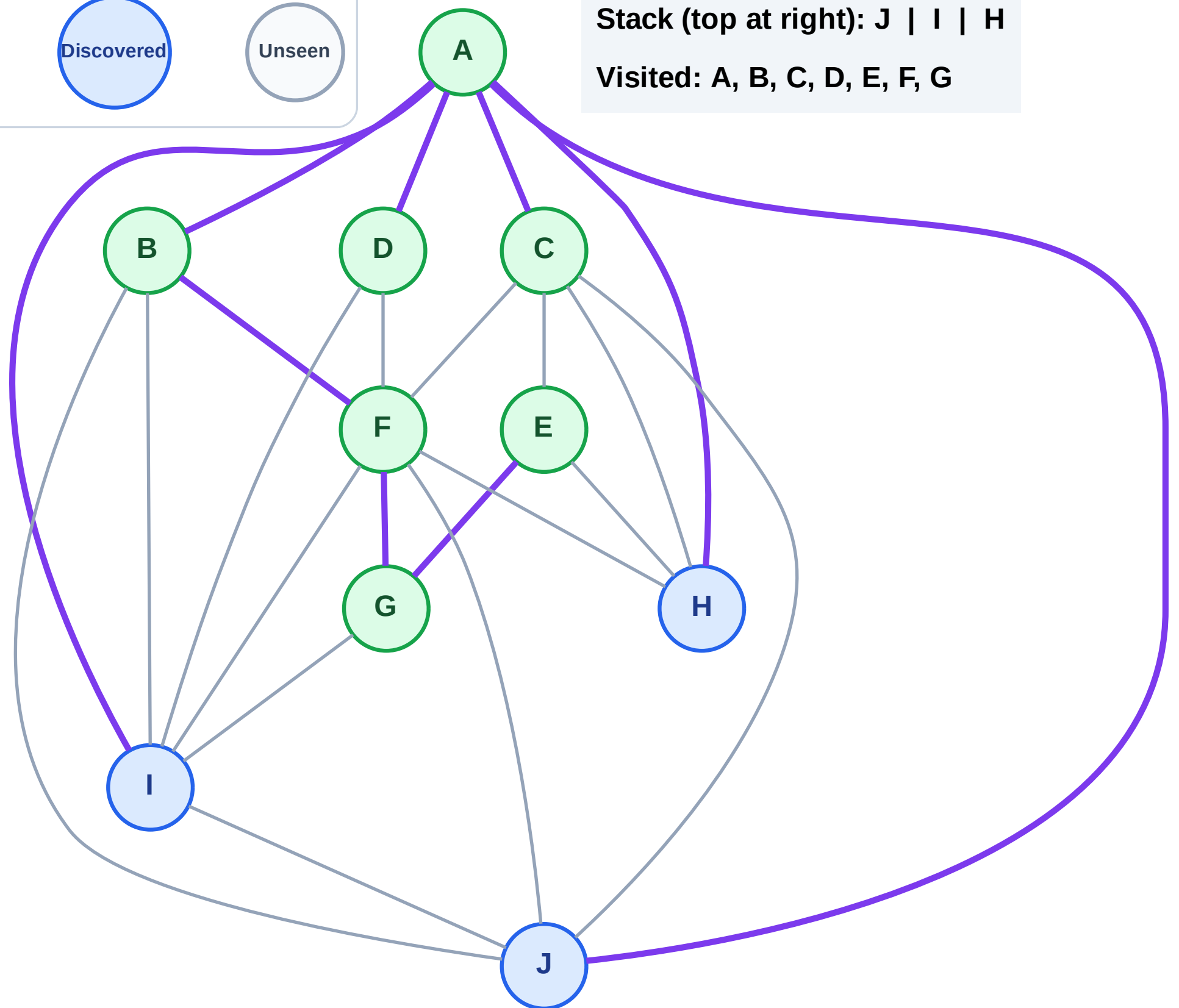
Step 15: No new neighbors from D

Legend



Stack (top at right): J | I | H

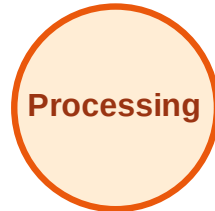
Visited: A, B, C, D, E, F, G



DFS Traversal

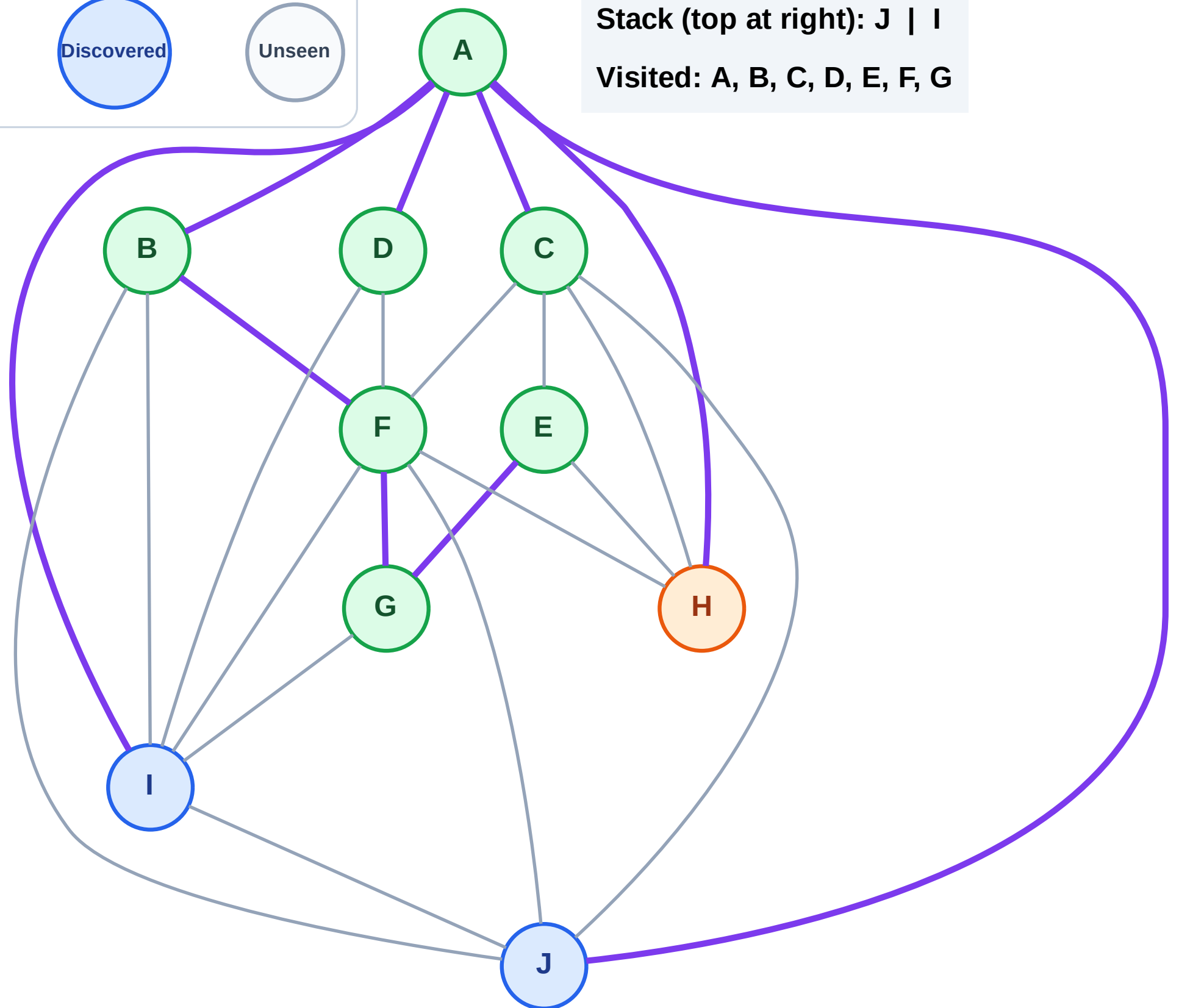
Step 16: Process node H

Legend



Stack (top at right): J | I

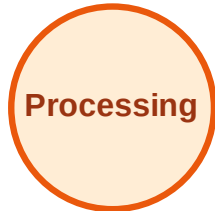
Visited: A, B, C, D, E, F, G



DFS Traversal

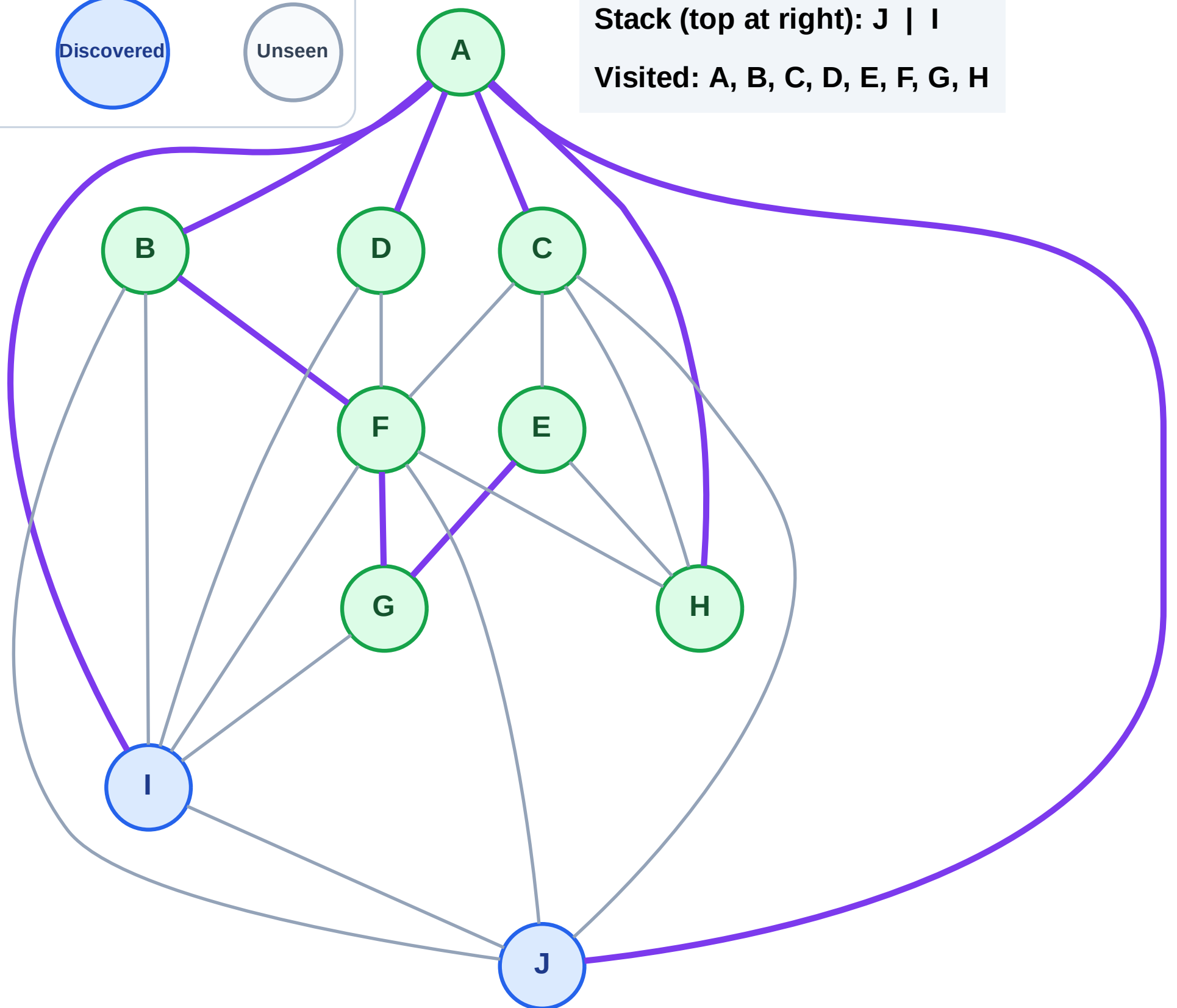
Step 17: No new neighbors from H

Legend



Stack (top at right): J | I

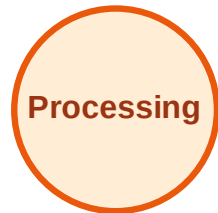
Visited: A, B, C, D, E, F, G, H



DFS Traversal

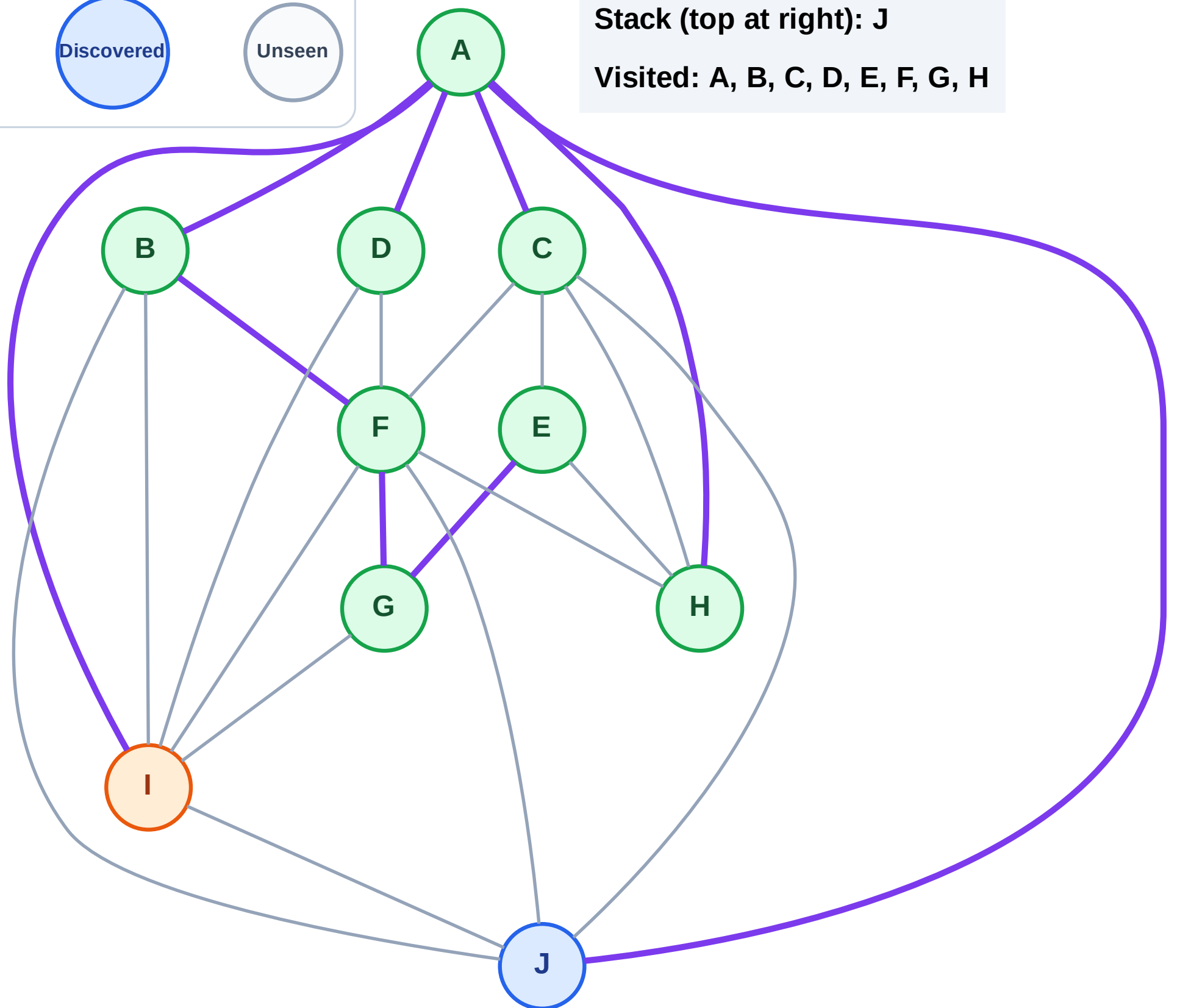
Step 18: Process node I

Legend



Stack (top at right): J

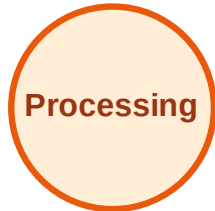
Visited: A, B, C, D, E, F, G, H



DFS Traversal

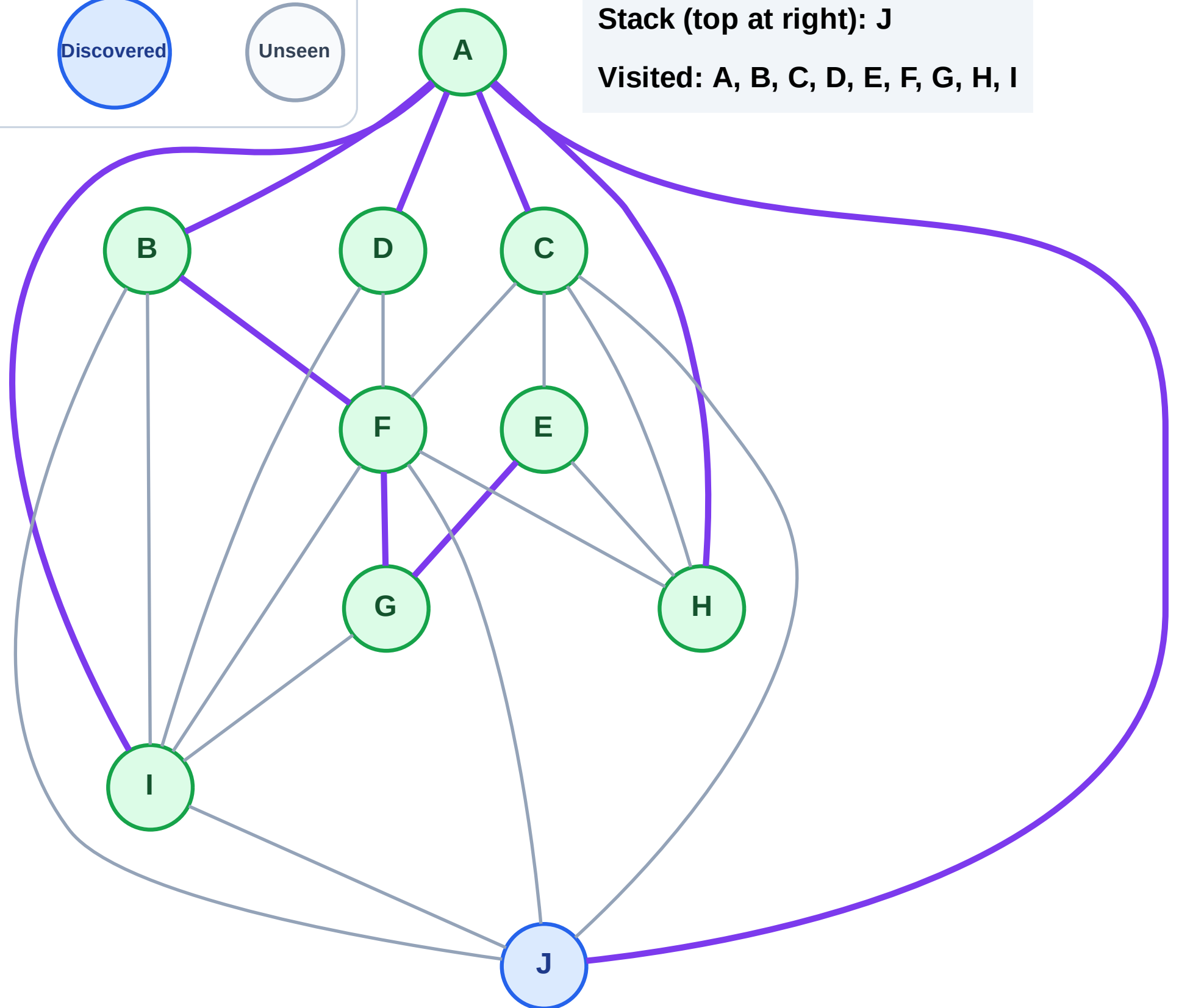
Step 19: No new neighbors from I

Legend



Stack (top at right): J

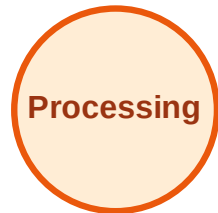
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

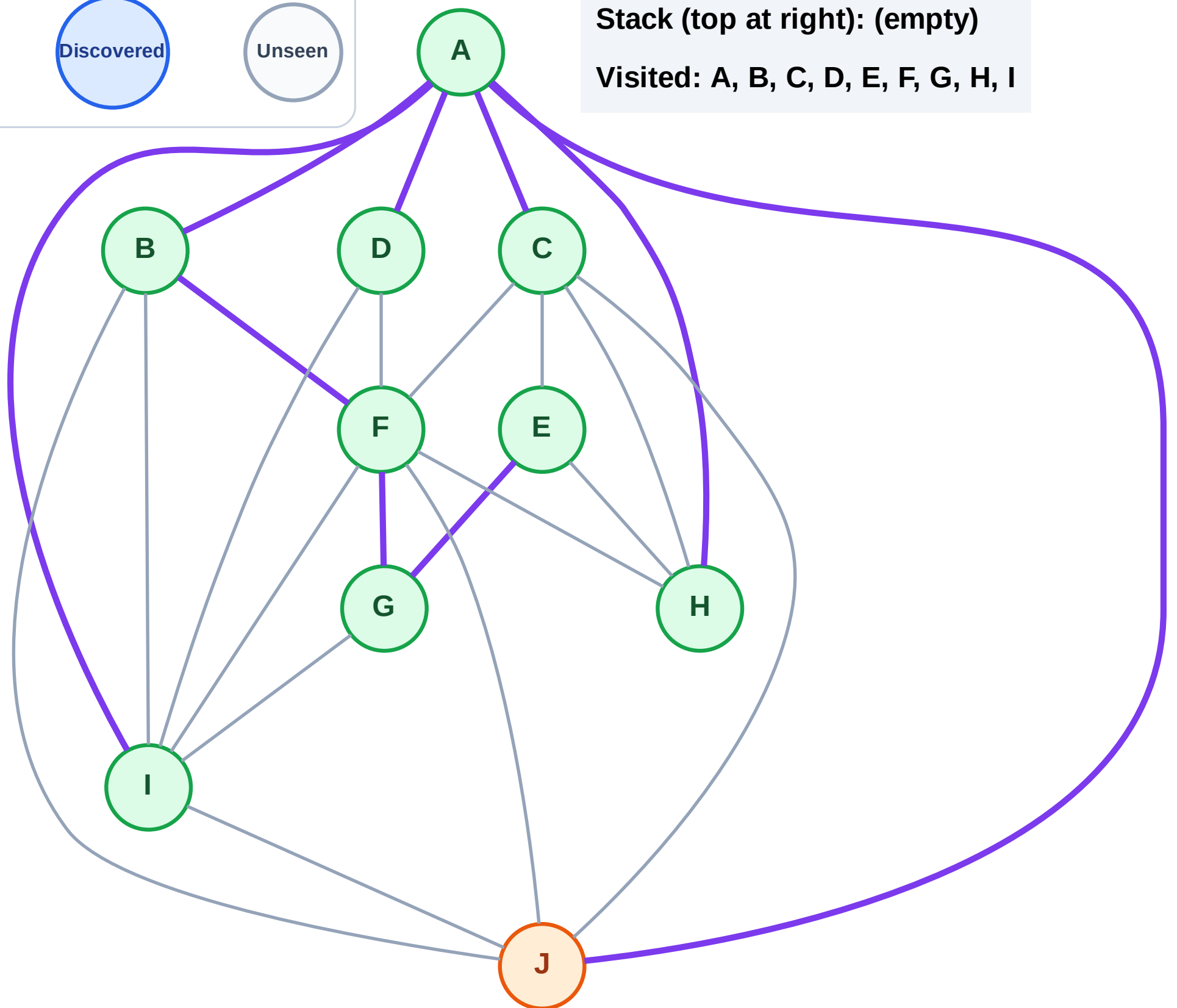
Step 20: Process node J

Legend



Stack (top at right): (empty)

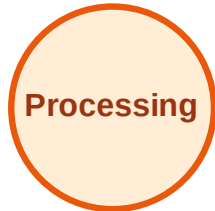
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

